



Procleix Reagent Equilibration System Model 177000GR Service Manual

SCREENING

This Service Manual was produced for use with the Procleix Reagent Equilibration System (RES).



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Revision History

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409793	1	N2400393	JUN 2022	Initial Release
409793	2	N2400393	JUL 2022	Appendix B removed, RES spare part list document created. Modified "Section 2. Carousel/Mounting Service Procedures" to clarify how to replace a clip. Modified part numbers of Chamber Wall Bearing and Drive Shaft O-Ring.
409793	3	N2400393	NOV 2022	Added the Carousel Removal Procedure in "Section 2. Carousel/Mounting Service Procedures". Modified Section 3: added Fuses information. Modified the number identification of the images to improve clarity.
CTS-SRV-SVD-000115 (409793)	4	N2400393	MAR 2024	Updated the Carousel Gear Motor PN from 927-0030 to B1907289.
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About This Manual

This Service Manual provides information regarding maintenance of the Procleix Reagent Equilibration System (RES).

SAP P/N	Vendor P/N	Description
741597	177000GR	Procleix Reagent Equilibration System

How This Manual Is Organized

This manual is divided into the following sections:

- General Procedures
- Specific Service Procedures for
 - Carousel/Mounting
 - Electronics Service
 - Carousel Drive
 - Heater/Temperature Sensor
- Appendices
 - Figures
 - List of Spare Parts
 - Contact Information

Highlighting and Quotation Conventions

Highlighting and quotation marks are used to convey specific meaning within the context in which they are used. The context and meaning of the highlighting and quotation marks are as defined below:

Italics are used for:

- Captioning under the figures displayed.

Bold is used for:

- Headings
- Text that needs to be delineated from surrounding text.

Special Symbols

The symbols below are used to warn of potentially dangerous situations and draw attention to other particularly important information.



Warning: Read these operating instructions fully before use and pay particular attention to sections containing this symbol.

Trademarks

The following terms are trademarks:

These terms...	Are trademarks of...
Procleix	Grifols Worldwide Operations Limited

Any other brand names in this manual are trademarks of their respective companies.

Related Documentation

The documents listed below contain additional information related to or referenced by topics described in this manual:

Title
<i>Procleix Reagent Equilibration System Operator's Manual</i>
Installation and Operational Qualification of the Procleix Reagent Equilibration System (RES) (409773)*

***Note:** Refer to 409773 for installation instructions for the Procleix Reagent Equilibration System.

Section 1. General Procedures

- Power Removal for Service
- Rear Cover
- Top Cover
- Warnings and Cautions

Power Removal for Service

Before any service procedure, turn power switch off and unplug power cord.

See Warnings and Cautions below.

Rear Cover

Most service operations require removal of the Rear Cover (Figure 1.1).

To remove cover, remove screws (8-32 x 3/8" Pan Head SEMS Phillips, quantity 18) from rear face of unit.

To replace cover, reverse process.

Top Cover

Some service operations require removal of the Top Cover (Figure 1.1). Additionally, some procedures which do not absolutely require it become easier with removal of this cover.

To remove cover:

- (Rear Cover must already be removed.)
- Unplug connectors on two wires from cooling fan. Note polarity indicated by wire color: Black to Black; Red to Red.
- Remove nuts across the top inside flange (8-32 KEPS, quantity 4) using size 11/32 (0.344) inch socket and wrench with a long (12 inch or more) extension.
- Slide cover ¼ inch [6mm] to the rear.
- Lift cover to disengage sheet metal tabs from slots on left and right sides of cover.

To replace cover, reverse process.

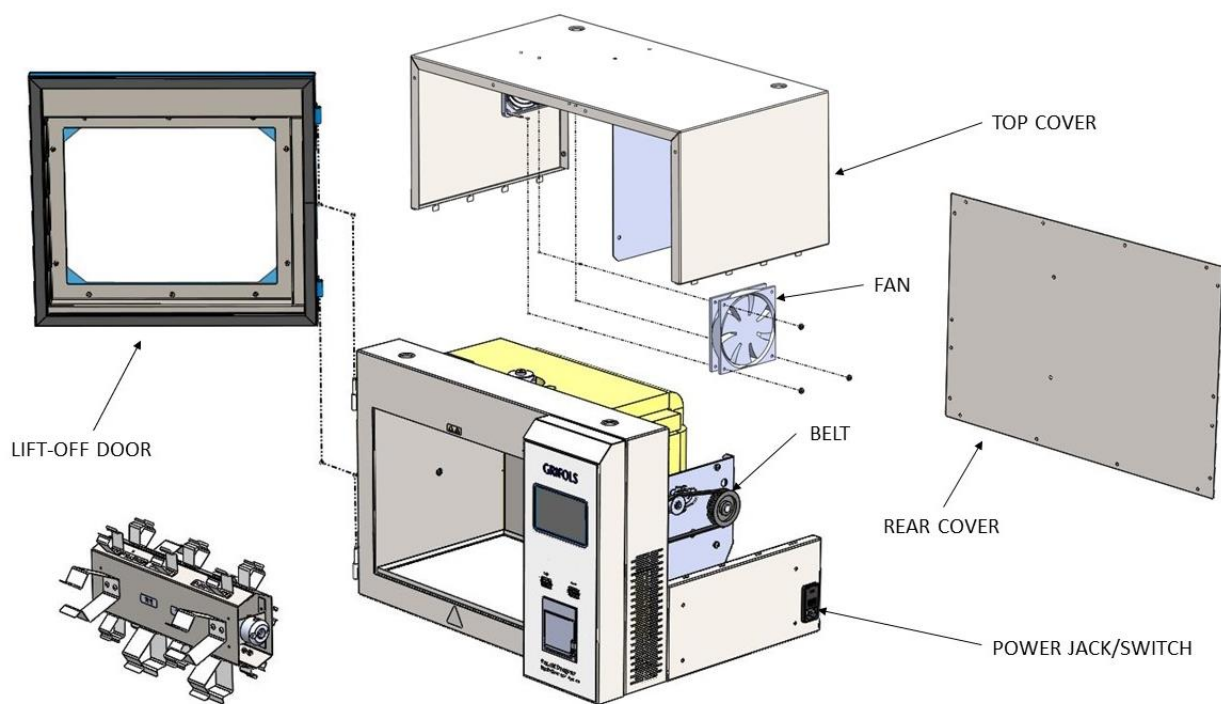


Figure 1.1. Major Features

Warnings and Cautions

Safety Notices

When cover/s are removed, there must be no power to unit, for these reasons:



- Live voltages are accessible, up to and including AC line voltage.
- Whenever the unit is powered, the enclosure fan runs, quietly and almost imperceptibly. This fan has no guard on the inside. (Figure 1.1). Thus, the cooling fan could cause injury when the rear cover is not present.
- The belt drive is unguarded (Figure 1.1) and could cause injury when cover/s are removed.

When moving the unit, be mindful that the door is mounted on lift-off hinges. (Figure 1.1)



- Do not lift or support unit under the door. It will separate from unit.
- Deliberately removing the door is a good way to reduce unit weight for easier handling. (Door assembly weighs 9 lb. [4 kg].)

When completing any service procedure that disturbs wiring, ensure that wiring is routed and dressed in original manner.



- Wiring must be isolated from moving parts of belt drive.
- Wiring should be supported without stress.
- Wiring should not encounter sharp metal edges.

Section 2. Carousel/Mounting Service Procedures

- Carousel
- Clip
- Ball Plunger
- Chamber Wall Bearing
- Drive Shaft O-Ring

Carousel (D1906923)

Carousel must be removed for any of the following service procedures.

Steps for carousel removal and replacement

1. Turn power switch to the OFF (O) position.
2. Remove main power supply cord.
3. Remove carousel from the chamber.
 - ▶ To remove, first rotate the carousel so that the large face of the six-sided carousel faces the door opening. This will orient the carousel for removal.
 - ▶ Insert the extended fingers of both hands between carousel and right-hand chamber wall, as shown in Figure 2.1, and push against the wall with both thumbs. Avoid using carousel clips as handles, because they may bend.

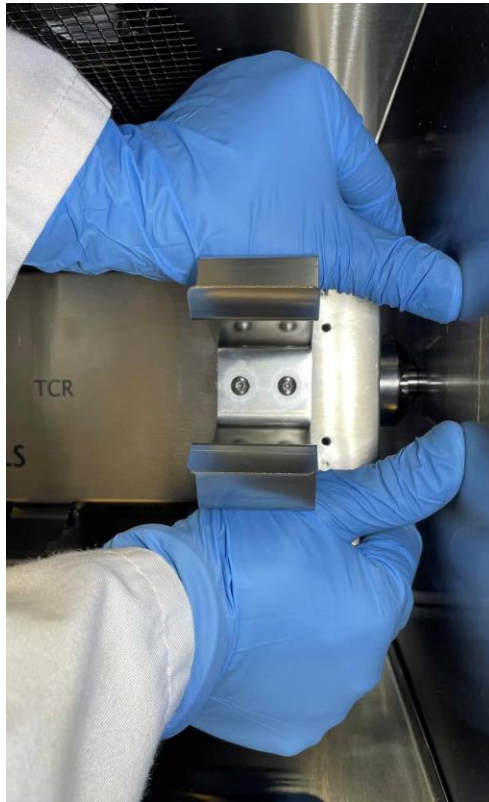


Figure 2.1 Carousel Removal by Pushing Left

- ▶ Gradually apply increasing force towards the left until the carousel disengages from the drive shaft.
 - ▶ Bring the right end of the carousel towards the door opening to clear the drive shaft.
 - ▶ Finally, move the carousel to the right to disengage the left end of shaft from the wall-mounted bearing, and remove the carousel from the machine. Inspect chamber for damage and corrosion and repair if necessary.
4. Re-install the carousel.
- ▶ To reinstall the carousel assembly, orient the carousel with the socket end to the right, and one of the large flat faces towards the user.
 - ▶ Angle the carousel's left end slightly into the chamber and insert the protruding shaft end into the wall-mounted bearing.
 - ▶ With the carousel inserted fully to the left, the socket can be aligned with the drive shaft protruding from the right.
 - ▶ Engage the socket with the drive shaft. Both shaft and socket have a triangular shape, so it may be necessary to rotate the carousel slightly for the shaft and socket to mate.
 - ▶ Once in position, the socket needs axial force to fully engage. This is most easily accomplished by inserting the extended fingers of both hands between carousel and left-hand wall, as shown in Figure

2.2, and pushing against the wall with both thumbs. A slight click might be felt when the socket is fully engaged.

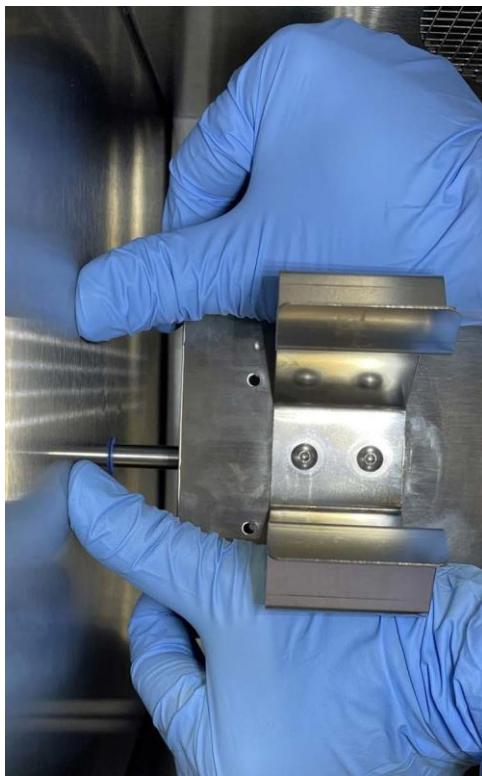


Figure 2.2 Carousel Insertion by Pushing Right

- Confirm full engagement by checking alignment of the mark on the drive shaft with the face of the socket, as shown in Figure 2.3 below.

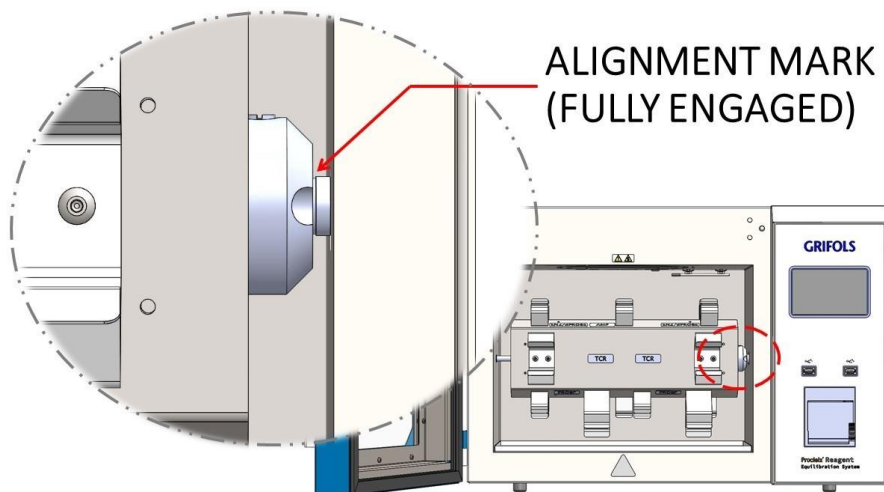


Figure 2.3 Alignment Mark

Clip (B190689...1/2/3)

Clips are located on all six faces of the carousel. (Figure 2.4) There are three sizes of clips:

- Small (B1906891)
- Medium (B1906892)
- Large (B1906893)

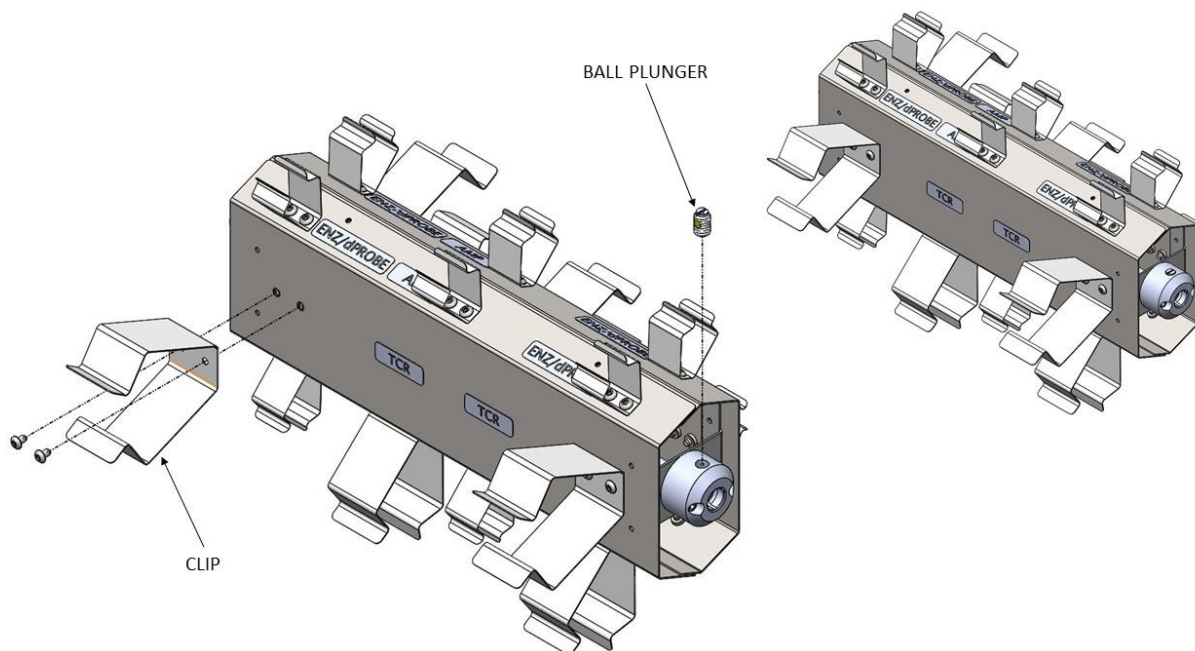


Figure 2.4. Carousel

To replace a clip:

- Remove carousel per the *Procleix Reagent Equilibration System Operator's Manual* and place on stable support. Avoid supporting carousel on clips, since these can easily become bent.
- For large clip and medium clip, remove screws (8-32 x 1/4" Button Head socket, quantity 2) from clip, using size 3/32 (0.094) inch hex wrench. Replace new clip of same size with screws.
- For small clip, remove screws (8-32 x 1/2" flat head socket, quantity 2) and spacer from clip. Using size 3/32 (0.094) inch hex wrench. Replace new clip of same size and new spacer with screws.
- Replace carousel per the *Procleix Reagent Equilibration System Operator's Manual*, current version.

Ball Plunger (922-0876)

The Ball Plunger is mounted in the Hub on right-hand end of the carousel. (Figure 2.4)

To replace Ball Plunger:

- Remove carousel and place on stable support. Avoid supporting carousel on clips, since these can easily become bent.
- Remove Ball Plunger with a large common flat screwdriver (nominal size 1/4"). Due to the built-in thread locking element, it will not spin freely. It requires effort to turn.
- Install new Ball Plunger. (Again, some effort is required.)

-
- Adjust depth of Plunger to produce desired mating with Drive Shaft upon Carousel replacement. Shaft should click into the Hub and be positively retained, but removable with several pounds of pull.

When Ball Plunger is replaced, simultaneous replacement of Drive Shaft O-Ring (922-0877) is recommended. In such case, it may be easier to determine ideal adjustment of Ball Plunger while the Drive Shaft O-Ring is not in place.

Chamber Wall Bearing (902-0157)

The chamber wall bearing is mounted in the left-hand chamber wall. (Figure 2.5)

To replace this bearing:

- Remove carousel.
- Pry old bearing out of hole using small needle nosed pliers.
- Install new bearing in hole.
 - Insert smaller end first.
 - Pinch the gap in the bearing's "C" shape to reduce diameter.
 - Spiral bearing into place with clockwise motion.
- Ensure flanged bearing is seated flat against wall.
- Replace carousel.

Drive Shaft O-Ring (922-0877)

The Drive Shaft O-ring is mounted on the drive shaft that protrudes from the right-hand chamber wall. (Figure 2.5)

To replace Drive Shaft O-ring:

- Remove carousel.
- Remove old Drive Shaft O-ring using a sharp probe (e.g., a pick tool or an unbent paper clip)
- Roll the new Drive Shaft O-ring over tip of shaft and into the groove.
- Ensure Drive Shaft O-ring is fully seated in the groove and therefore cannot roll out.
- Replace carousel.

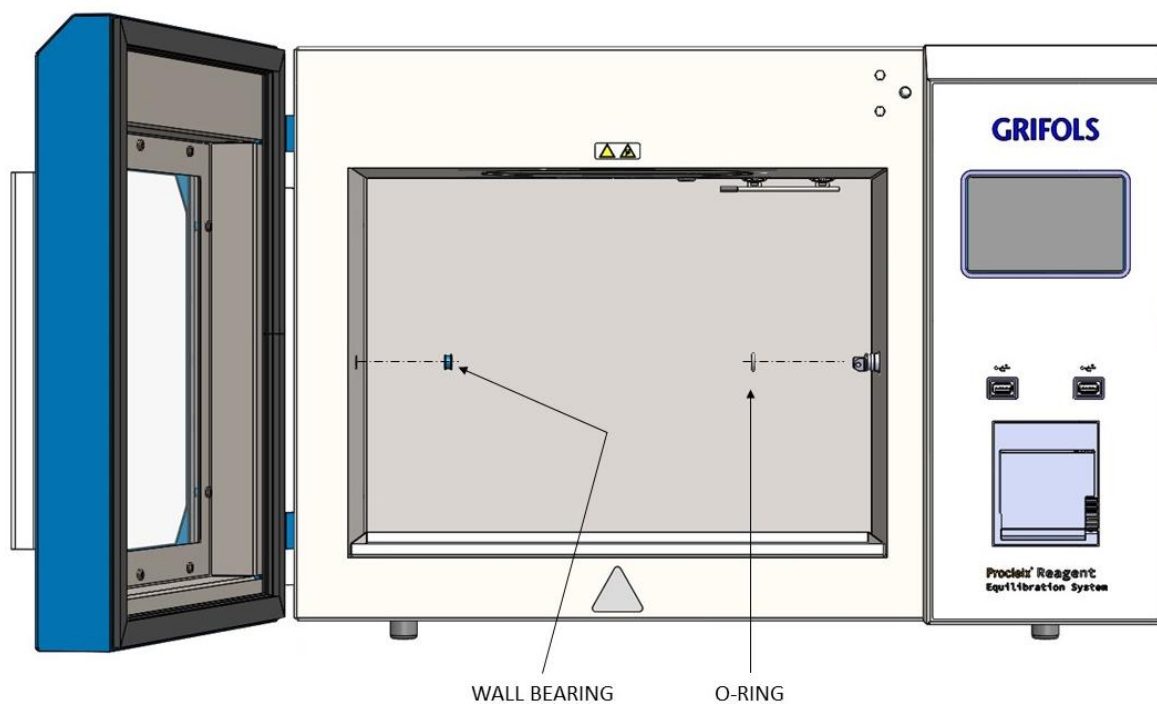


Figure 2.5. Carousel Mounting Parts

Section 3. Electronics Service

- General Guidance
- Thermal Printer
- Touchscreen/Display Board
- Main Board
- Power Supply
- Enclosure Fan
- Power Entry Module
- Fuses

General Guidance



- Observe static precautions when handling electronic components, especially circuit boards.
- Except for Thermal Printer, all electronic components require removal of the Rear Cover for service.

Thermal Printer (902-1062)

The Thermal Printer is located on the Front Panel.

To remove Printer:

- Remove paper from Printer. Leave paper door open.
- Two screws are visible in the paper holding cavity. (Screws are self-retaining and will not fall out when loosened.). Loosen both completely using #2 Phillips screwdriver.
- Printer is not free yet - a cable is attached to the rear. Carefully pull Printer out of panel just until the connector becomes accessible.
- Remove connector from Printer.

To replace Printer, reverse process.

Touchscreen / Display Board (916-0069)

The Touchscreen / Display Board is mounted against inside of the Front Panel. (Figure 4) The LCD display is an inseparable part of the board assembly.

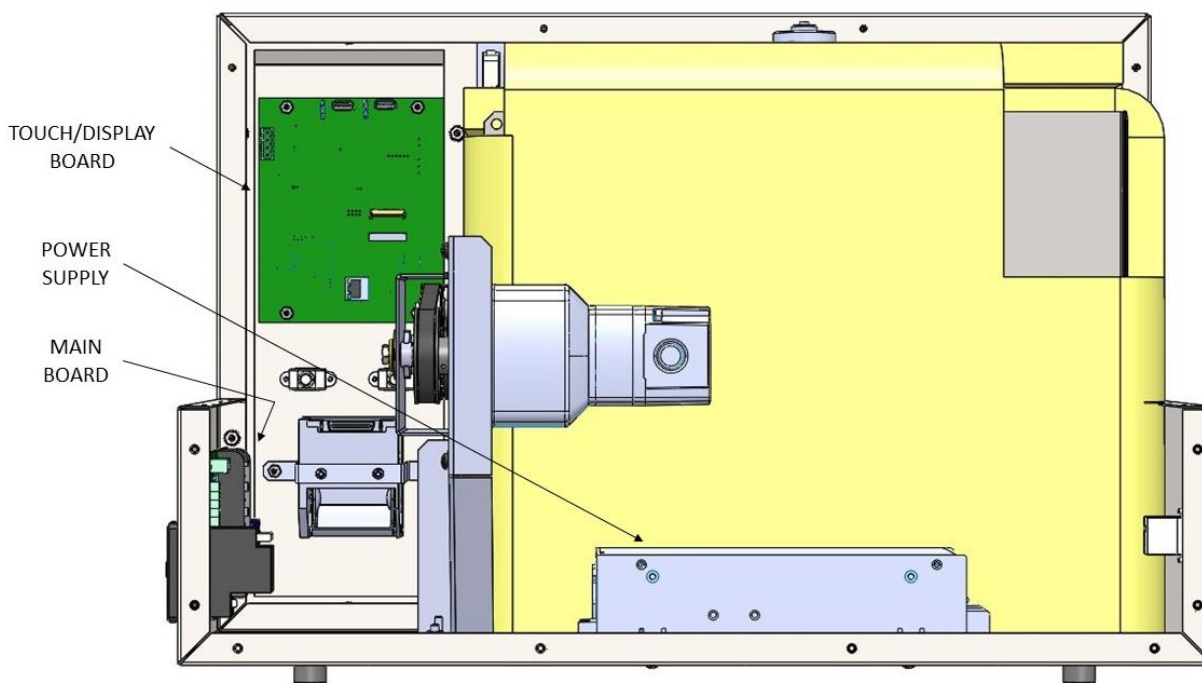


Figure 3.1. Electronics Components Locations (Rear View)

To remove Board:

- Remove Rear Cover. (Top Cover removal is optional but recommended.)
- Disconnect all cables from Board. (Figure 3.2)

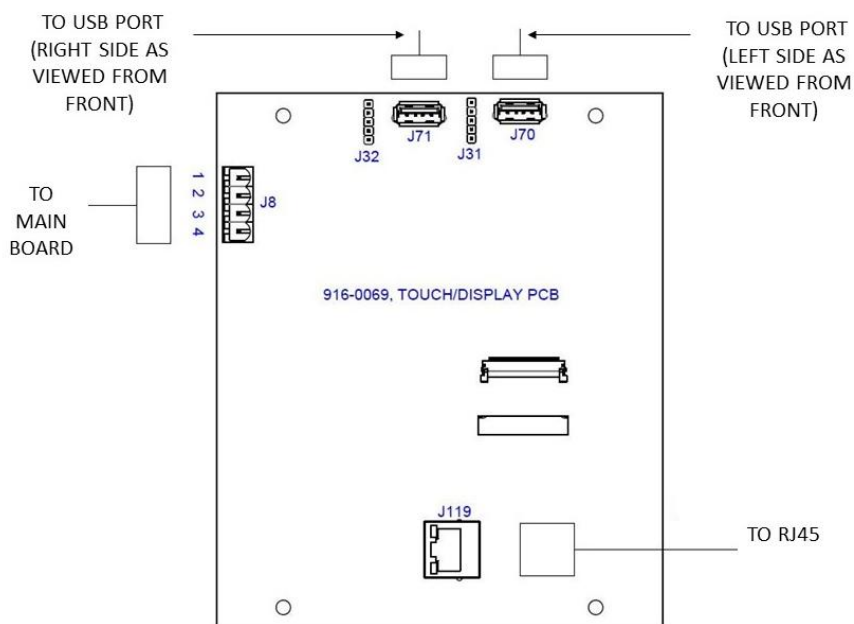


Figure 3.2. Display Board Connections

- Remove nuts at each corner of Board (6-32 KEPS, quantity 4) using size 5/16 (0.313) inch socket and wrench. A long (12 inch or more) extension is helpful.
- Carefully remove Board, leaving white plastic spacers in place at each mounting location.

To replace Board:

- If Board is new, peel clear protective plastic film from face of LCD.
- Check that white plastic spacer is in place at each of four mounting locations.
- Carefully mount Board on four threaded studs. Ensure Board orientation is correct: Twin USB connectors should be at the top.
- Replace four nuts.
- Reconnect all cables. (Figure 3.2)
- Replace cover/s.

Main Board (916-0070)

The Main Board is mounted against inside of the enclosure side wall. (Figure 3.3)

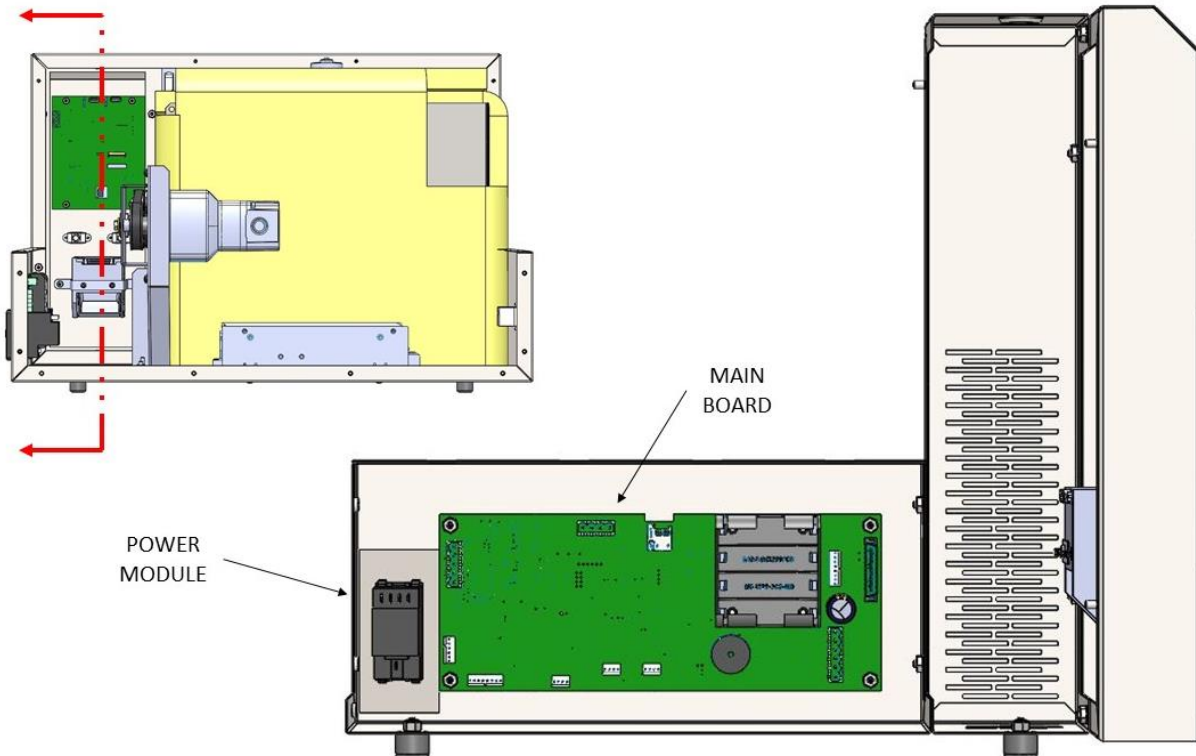


Figure 3.3. Electronics Components Locations (Side Section View)

To remove Board:

- Remove Rear Cover. (Top Cover removal is optional but recommended.)
- Disconnect all cables from Board. (Figure 3.4)

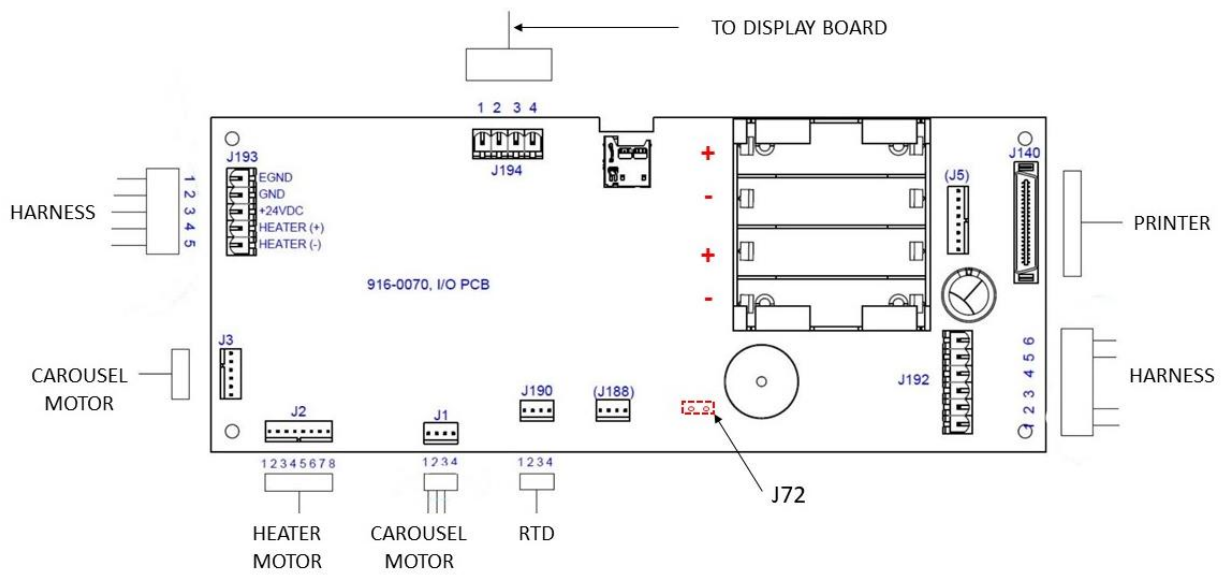


Figure 3.4. Main Board Connections

- Remove Batteries from Board.
- Remove nuts at each corner of Board (6-32 KEPS, quantity 4) using size 5/16 (0.313) inch socket and wrench.
- Carefully remove Board, leaving white plastic spacers in place at each mounting location.

To replace Board:

- Inspect new Board. Ensure that any jumper has been removed from J72. (Figure 3.4)
- Check that white plastic spacer is in place at each of four mounting locations.
- Carefully mount Board on four threaded studs. Ensure Board orientation is correct: Battery holder should be towards front of the unit.
- Replace four nuts.
- Replace batteries, noting polarity. (Figure 3.4)
- Reconnect all cables. (Figure 3.4)
- Replace cover/s.

Power Supply (902-1088)

The Power Supply is mounted just inside the Rear Cover, on the enclosure floor. (Figure 3.1) Mounting screws are accessed from beneath the unit, so it is necessary to tip the unit. These measures are recommended:

- Get assistance from a second person.
- Prevent Door from opening while tipping the unit.
- Be careful not to lift the Door off its hinges.

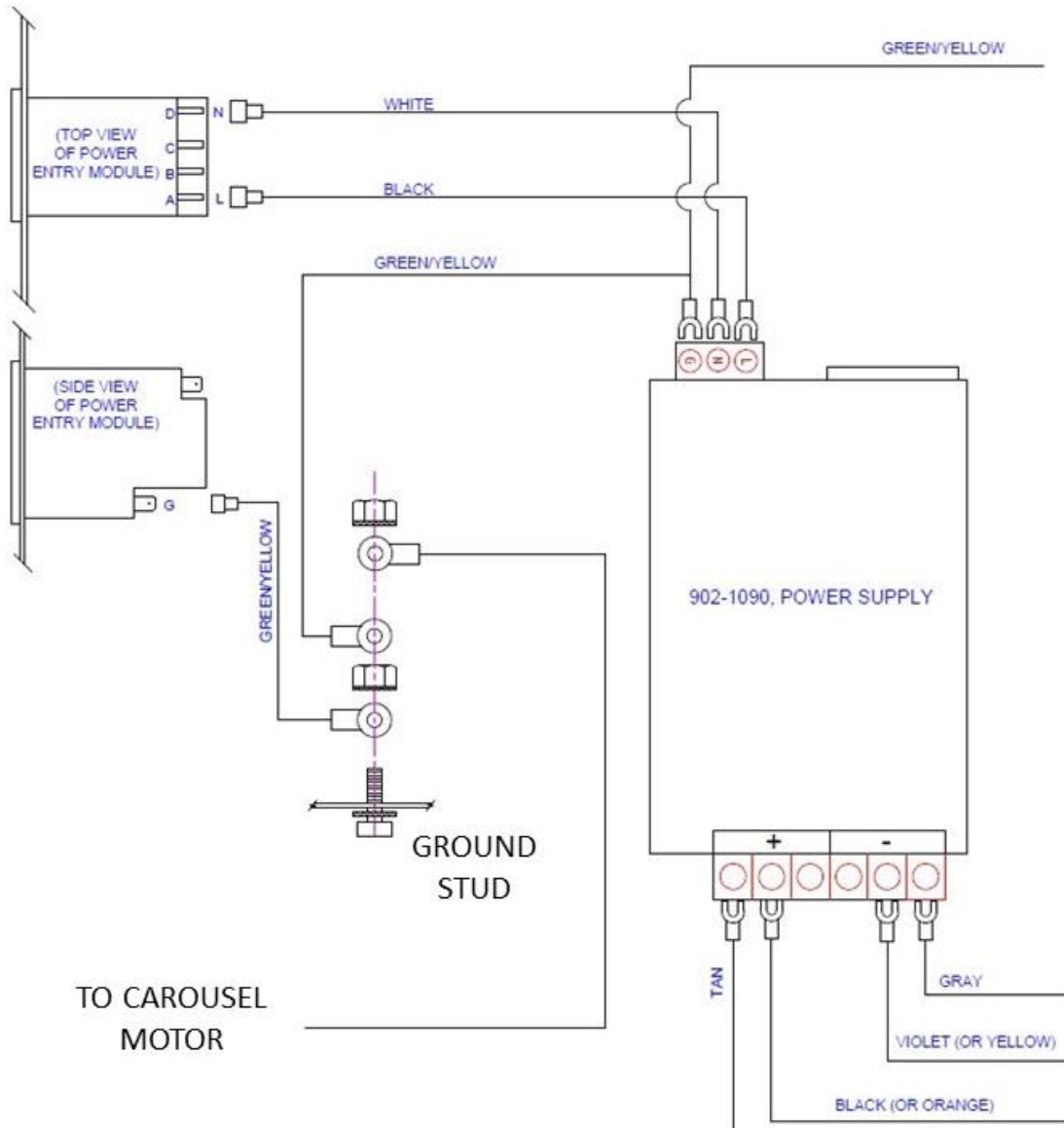


Figure 3.5. Power Module and Supply Connections

To remove Power Supply:

- Remove Rear Cover. (Top Cover removal is optional.)
- Remove clear plastic covers from Power Supply terminals.
- Mark locations for all wiring before disconnecting, to facilitate reconnection.
- Disconnect all wiring from Power Supply (Figure 3.5) using a short length #1 or #2 Phillips screwdriver.
- Tip unit as recommended above.
- Remove four screws (M4 x 6mm Pan Head Phillips, quantity 4) with externally toothed lock washers (quantity 4) from underneath unit.
- Remove Power Supply.

To replace Power Supply, reverse process.

Enclosure Fan (902-1156)

To remove Fan:

- (Rear Cover must already be removed.)
- Unplug connectors on two wires from Fan. Note polarity indicated by wire color: Black to Black; Red to Red.
- Remove nuts (6-32 KEPS, quantity 3) using size 5/16 (0.313) inch socket and wrench. (Figure 1.1)
- Remove Fan, noting orientation: wires exit fan towards front of unit.
- Leave Fan Guard in place.

To replace Fan:

- Check that Fan Guard is still in place.
- Find flow direction arrow on Fan body. This must point out of unit.
- With flow direction arrow pointing out of unit, orient Fan so that wiring exits the Fan toward front of unit (farthest from Rear Cover).
- Install Fan with nuts.
- Reconnect two wires according to polarity indicated by wire color: Black to Black; Red to Red.
- Replace cover/s.
- After unit is operational, verify airflow exhausts from unit at fan.

Power Entry Module (902-1089)

The Power Entry Module is mounted in rear corner of the unit. (Figure 3.3) The external power switch is part of the Module.)

The Module is snapped into a rectangular hole in the enclosure Base. (A loose sheet metal shim, with identical rectangular hole, is used to increase the total wall thickness at this location.) No threaded fasteners are used.

To remove Module:

- Remove Rear Cover. (Top Cover removal is not needed.)

-
- Mark locations for wiring before disconnecting, to facilitate reconnection.
 - Disconnect wiring from Module. (Figure 3.5)
 - Press down on two snap fingers molded into the top of the Module, and tip Module out of the rectangular hole in enclosure Base.
 - Remove Module.
 - The Shim will fall inside the unit.

To replace Module:

- Hold Shim against inside wall, aligning rectangular holes.
- With other hand, orient the Module with cable jack at the bottom.
- Snap Module into hole.
- Ensure that Module is fully seated and firmly retained by Shim.
- Reconnect wiring. (Figure 3.5)
- Replace cover.

Fuses (902-1103)

There are two fuses located in the fuse drawer on the power entry module.

To remove fuses:

- Turn power switch to the OFF (O) position.
- Disconnect the power cord from mains voltage.
- Remove the power cord from the power entry module.
- On the power entry module, open the hinged access cover to the fuse drawer.
- Pull back on and remove the fuse drawer. (The use of a small flathead screwdriver will assist.)
- Take out the fuses.

To replace fuses:

- Replace with the correct fuses. The fuses should be 5 mm x 20 mm fast acting, rated 10 A, 250 V.
- Push the fuse drawer back in and close hinged access cover.
- Reconnect power cord to power entry module on one end and to mains voltage on the other.
- Return power switch to the 'ON' (I) position.

Section 4. Carousel Drive Service

- General Guidance
- Tensioner
- Carousel Gear Motor
- Timing Belt
- Drive Shaft Bearings

General Guidance

When installed, the removable Carousel is coupled with a fixed Drive Shaft. Pulleys are mounted on this Drive Shaft and the Drive Motor, and the two Pulleys are connected by a Timing Belt. A spring-loaded Tensioner is used to keep the Belt tight. (Figure 4.1 and Figure 4.2)

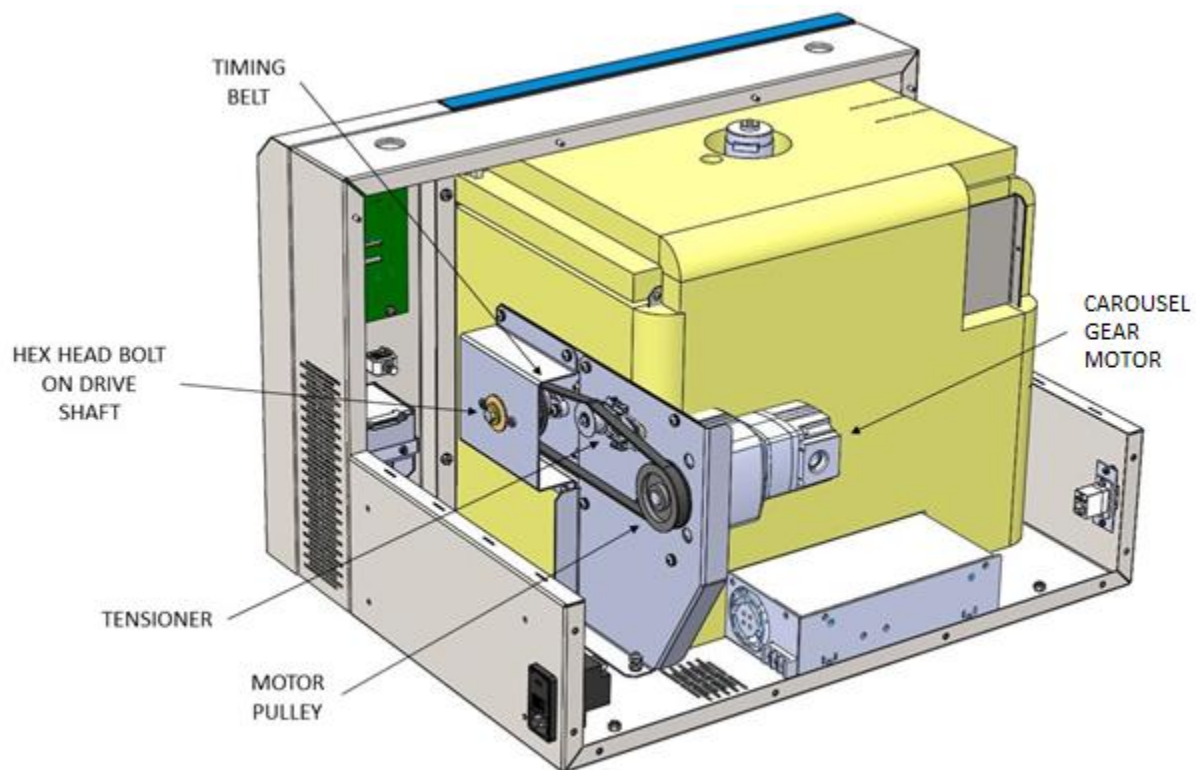


Figure 4.1. Carousel Drive Parts

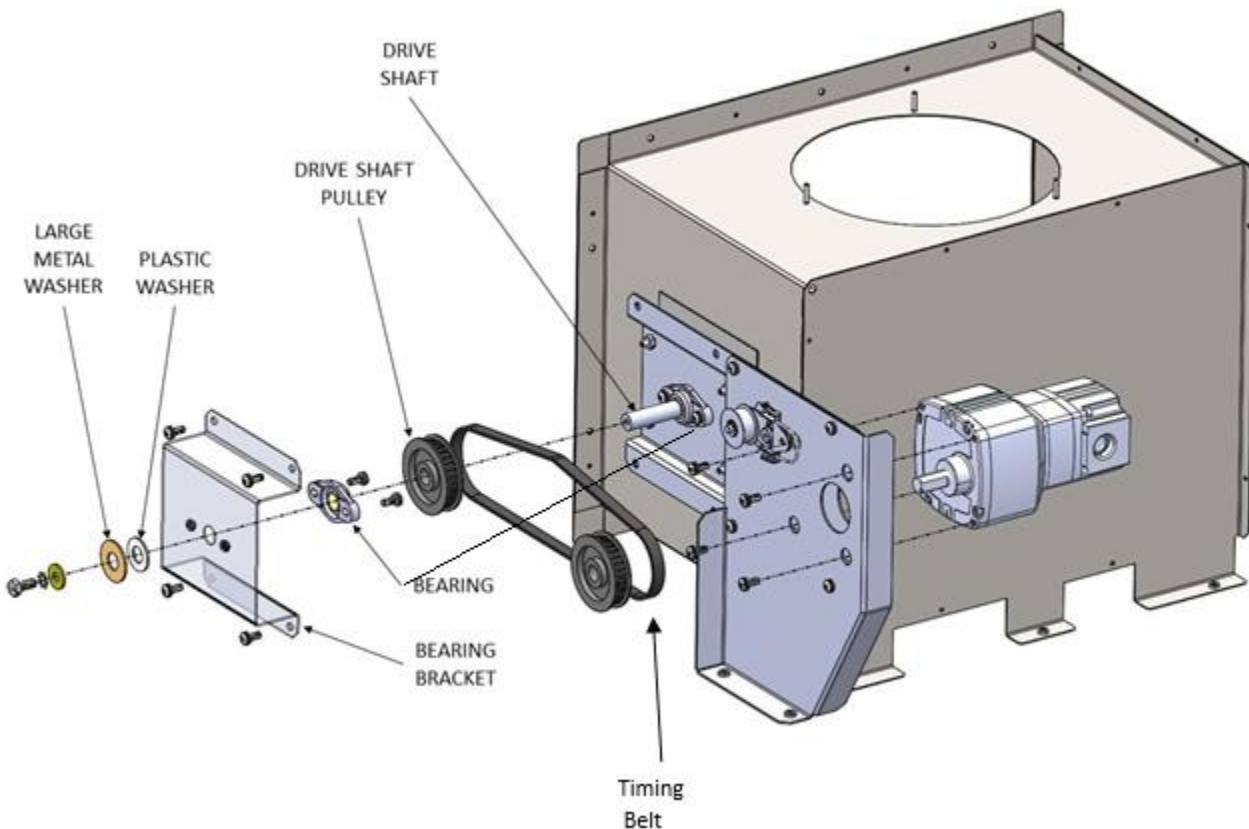


Figure 4.2. Carousel Drive (Exploded View)

For all service operations to the Carousel Drive:

- Removal of both Rear and Top Covers is recommended.
- Loosening the Belt is required; then the final step is tightening the Belt again. These steps are described below under Tensioner.

Tensioner (902-1169)

- Tensioner Description (Figure 4.3)
 - The Tensioner is mounted by two features:
 1. a pin on back of the Tensioner fits into a hole in a Bracket on the RES unit, and
 2. a screw (with special shouldered “Follower” washer) clamps the Tensioner to the Bracket.
 - When the screw is loosened, the Tensioner can pivot on its pin within the range allowed by the screw in its slot.
 - The Tensioner consists of two “layers”:
 1. a Mounting Plate which mounts against the Bracket, and

2. a Carriage which holds the Pulley wheel.

- Both the Mounting Plate and the Carriage are mounted the same way (described above) and can move relative to each other. There is also a horseshoe-shaped Spring trapped between the two. This Spring pushes the Carriage in clockwise direction relative to the Mounting Plate.

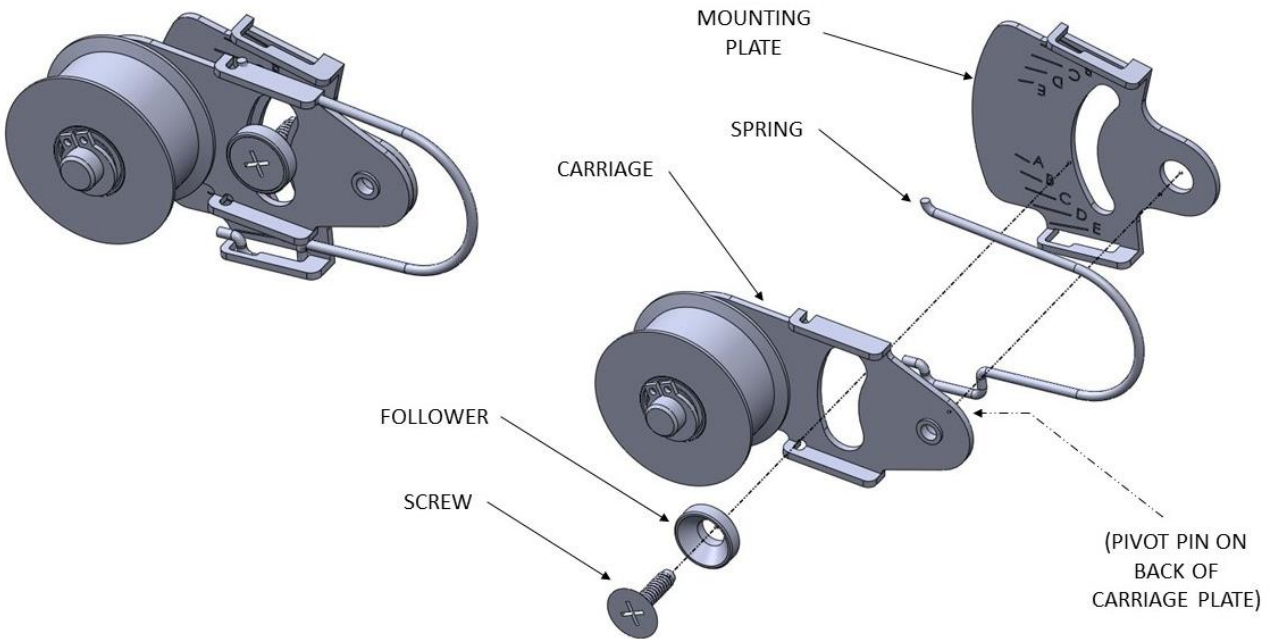
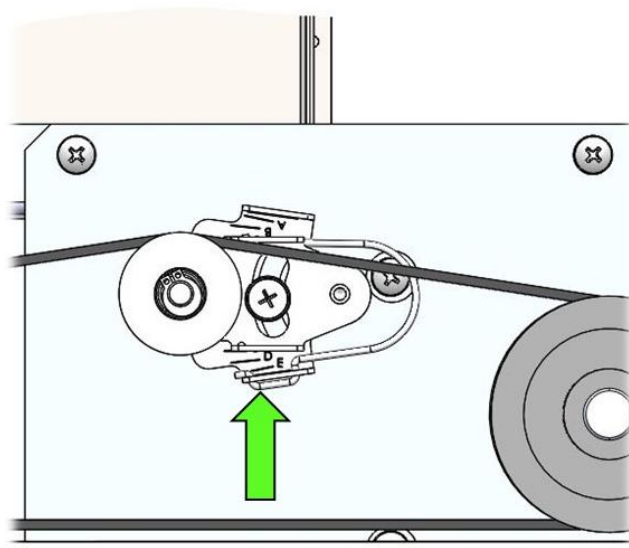
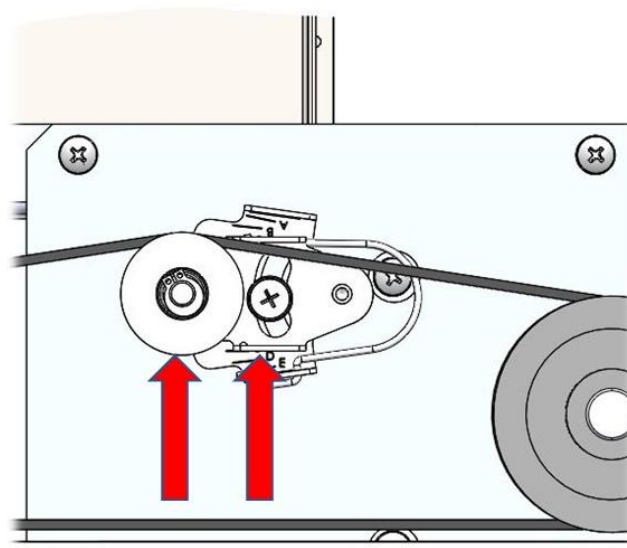


Figure 4.3. Tensioner Details

- Loosening Belt Tension
 - Remove covers.
 - Loosen the single screw on the Tensioner. (Do not remove the screw. This and the Follower washer are not standard.) The Tensioner will drop away from the Belt. (Figure 4.6)
 - Slip Belt off the free side of Tensioner Pulley and raise the Pulley above the Belt. This loosens the Belt a bit more.
- Tightening Belt Tension
 - Using forefinger of left hand, push up on the bottom of Mounting Plate. (Figure 4.5) This compresses the Spring and pushes the Carriage up against the Belt.
 - Push the Plate up as far as it can go, fully compressing the Spring. (Figure 4.6)
 - Tighten the screw while Plate is in this position.
 - Inspect the Tensioner. There are graduated marks on the Plate with letters to indicate degrees of Spring compression. (Disregard the set of letters that is upside down.) The Tensioner should be set to "E" (maximum).
 - Check again that the screw is firmly tightened. (Recommended torque 32 in.lb. [360 N-cm])
 - Replace covers.

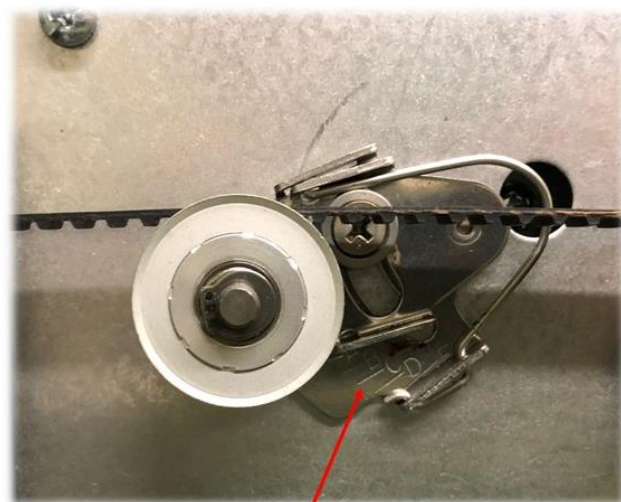


**PUSH UP HERE
(ON MOUNTING PLATE)**



**DO NOT PUSH UP HERE
(ON CARRIAGE OR PULLEY)**

Figure 4.5. Tensioner Push Point



SPRING IS FULLY RELAXED

FULLY LOOSENED TENSIONER



SPRING IS FULLY LOADED

FULLY TIGHTENED TENSIONER

Figure 4.6. Tensioner States

Carousel Gear Motor (B1907289)

To remove Motor (Figure 4.2):

- Remove covers.
- Loosen Belt tension (as described under "Tensioner")
- Turn Motor Pulley until set screw faces rear. Loosen set screw on Motor Pulley using size 3/32 (0.094) inch hex wrench. Do not remove set screw.
- Remove Pulley from Motor. Simultaneously disengage Belt from Pulley.
- Disconnect all wiring to Motor. (Figure 3.4 and Figure 3.5)
- Remove screws (10-32 x 3/8" Pan Head SEMS Phillips, quantity 4) mounting Motor to Bracket. Support Motor weight before removing final screw.
- Remove Motor.

To replace Motor (Figure 4.2):

- Install Motor in Bracket with (four) screws. Tighten screws securely.
- Turn Motor shaft by hand until flat face on shaft faces rear of unit.
- Engage Belt onto Pulley and install on Motor shaft. Turn Pulley so that set screw will land on flat face of Motor shaft.
- From behind unit, visually sight down the length of Belt. Slide Pulley in or out on Motor shaft until it is aligned laterally with other Pulleys on Tensioner and Drive Shaft. Tighten set screw using size 3/32 (0.094) inch hex wrench.
- Reconnect all wiring to Motor. (Figure 3.4 and Figure 3.5)
- Tighten Belt tension.
- Replace covers.

Timing Belt (902-1168)

To remove Belt (Figure 4.2):

- Before loosening Belt tension, loosen Hex Head Screw on end of Drive Shaft: Standing behind unit, grasp Motor Pulley tightly with left hand to immobilize Drive Shaft. Simultaneously loosen Screw with right hand, using 7/16 (.438) inch wrench. Alternatively, an assistant can immobilize the Carousel from in front of unit while the Screw is loosened. (Avoid bending clips.)
- Remove Carousel if present.
- Loosen Belt tension.
- Remove fasteners (this order) from end of Drive Shaft:
 - Screw (1/4-20 x 5/8" Hex Head)
 - Toothed Washer (1/4" External Tooth)
 - small metal Flat Washer (1/4")
 - large metal Flat Washer (1/2")
 - Plastic Washer (1/2")
- Remove screws (10-32 x 3/8" Pan Head SEMS Phillips, quantity 4) mounting Bearing Bracket and pull Bracket off the Drive Shaft.
- Remove Belt, rolling it off from Drive Shaft Pulley. (If this is too difficult, simultaneously remove Pulley from Motor as described under Motor removal. Do not remove Drive Shaft Pulley.)

To replace Belt (Figure 4.2):

- Install Belt, rolling it onto Drive Shaft Pullet. (If Motor Pulley was removed, replace it as described for Motor replacement.)
- Replace Bearing Bracket on Drive Shaft. Install with (four) screws.
- Replace fasteners on end of Drive Shaft in this order:
 - Plastic Washer
 - large metal Flat Washer
 - small metal Flat Washer
 - Toothed Washer
 - Screw (partially tightened)
- Tighten Belt tension (Figure 4.5 and Figure 4.6)
- Tighten Hex Head Screw securely, using same method as when removed.
- Check alignment of Pulleys as described for Motor replacement.
- Turn Motor Pulley by hand. Check for free, smooth operation of Belt drive.
- Replace covers.

Drive Shaft Bearings (A1905982)

There are two identical Drive Shaft Bearings. (Figure 4.2) These should be replaced as a pair (quantity = 2).

To remove Bearings:

- Complete Timing Belt removal procedure, as described above.
- Install Carousel in RES unit to stabilize Drive Shaft.
- Remove Pulley from Drive Shaft using size 3/32 (0.094) inch hex wrench. Do not remove set screw. Drive Shaft will remain in place.
- Remove screws (10-32 x 3/8" Pan Head SEMS Phillips, quantity 2) mounting (inboard) Bearing.
- Slip Bearing off Drive Shaft.
- Remove screws (10-32 x 3/8" Pan Head SEMS Phillips, quantity 2) mounting (outboard) Bearing to Bearing Bracket.

To replace Bearings:

- Slip new Bearing onto Drive Shaft and install with (two) screws.
- Replace Drive Shaft Pulley. If there is a flat face on drive shaft, rotate pulley such that set screw will clamp upon the flat face. Align this pulley with Motor and/or tensioner Pulleys so that Belt will run flat. Tighten set screw on Drive Shaft Pulley.
- Install new Bearing onto Bearing Bracket with (two) screws.
- Complete Timing Belt replacement procedure, as described above.

Section 5. Heater / Temperature Sensor Service

- General Guidance / Heater Access
- Heater Removal and Replacement
- Heater Impeller Motor
- Heat Element
- RTD Temperature Sensor
- Over-Temperature Switch

General Guidance / Heater Access

The heater consists of a circular resistance Element, a coaxial Impeller with Motor, and an Over-Temperature Switch, all mounted on a metal Pan.

These preliminary steps are required to access the Heater or Sensor for any service operation:

- Remove covers.
- A layer of insulation is folded over top of chamber and tucked in beneath overhang of enclosure front end. Retract this layer and allow it to stand upright to expose the layer beneath. (Figure 5.1)
- A second layer is folded over top of the chamber from the right side. Also pull this back and un/fold out of the way.

Now the Heater is accessible.

To re-cover the Heater after service, reverse this process.

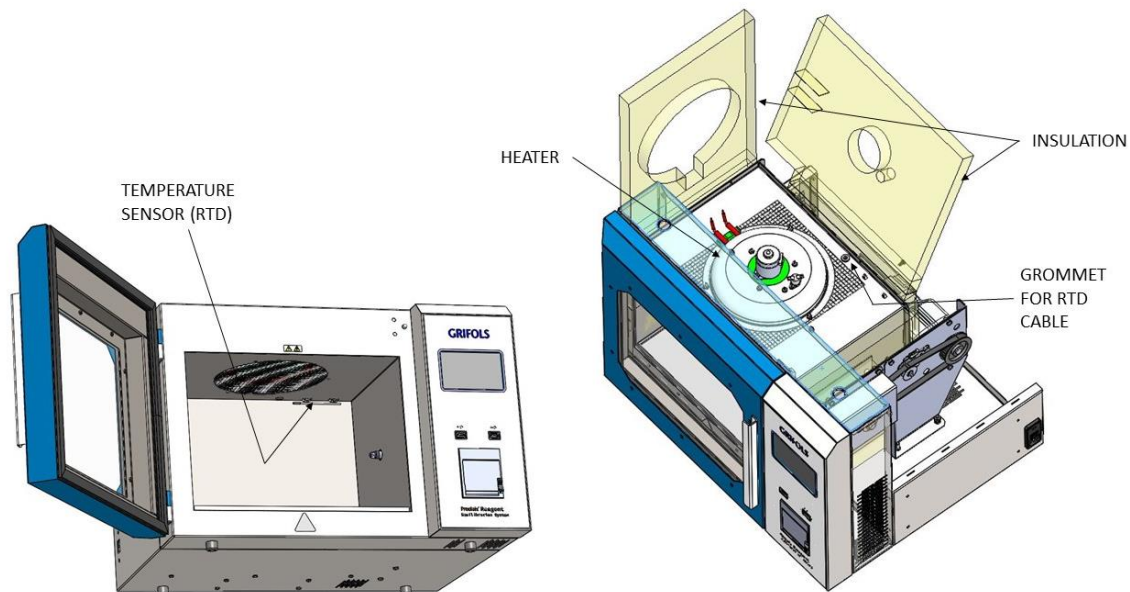


Figure 5.1. Heater Access

Heater Removal and Replacement

Some service procedures require removal of the entire Heater from unit.

To remove Heater:

- Access the Heater, as described above.
- Disconnect all wiring. Use special care when disconnecting the very small connector on Heater Impeller Motor: Do not pull small wires. Instead, use a small sharp object to pry upon connector at right and left ends, alternating ends. Do not damage wires.
- Remove nuts (6-32 KEPS, quantity 3) from perimeter of Heater Pan using size 5/16 (0.313) inch socket and wrench.
- Lift Heater off studs on Chamber.

Handle the Heater with care to avoid bending the Impeller. (A slightly bent Impeller can be very noisy when operating.)

To replace Heater:

- Mount Heater on studs atop Chamber. Note orientation: Heater Element wires exit toward the hinge side of the Chamber Door.
- Reinstall (three) nuts using size 5/16 (0.313) inch socket and wrench.
- Reconnect all wiring.
- Re-cover the Heater as described above.

Heater Impeller Motor (927-0031)

To remove Heater Impeller Motor with Impeller as Subassembly (Figure 5.2):

- Remove the Heater, as described above.
- Remove nuts (6-32 KEPS, quantity 3) holding Motor Mounting Plate to Pan using size 5/16 (0.313) inch wrench. Support Impeller while removing final nut.
- The Motor, Impeller and two plates in between are still joined as a subassembly, now free from the Pan. Be careful not to bend Impeller.

To separate Heater Impeller Motor from Impeller (Figure 5.2):

- Loosen set screw on Impeller, using size 3/32 (0.094) inch hex wrench. Do not remove set screw.
- Remove the nuts (6-32 KEPS, quantity 2) holding Motor to Mounting Plate using size 5/16 (0.313) inch wrench.
- Separate Motor from Impeller plates. Be careful not to bend Impeller.

To rejoin Heater Impeller Motor with Impeller (Figure 5.2):

- Mount Insulator Plate on Mounting Plate with no fasteners.
- Mount Motor on Insulator Plate. Orient with connector towards flat side of Insulator. Install with (two) nuts.
- Mount Impeller on Motor shaft.
 - Location of Impeller on shaft is critical.
 - Set location on Shaft so that only tip (.04±.02 inch [1±0.5mm]) length is exposed beyond Impeller hub. (Figure 5.3)
 - Tighten set screw on Impeller.

- Be careful not to bend Impeller.

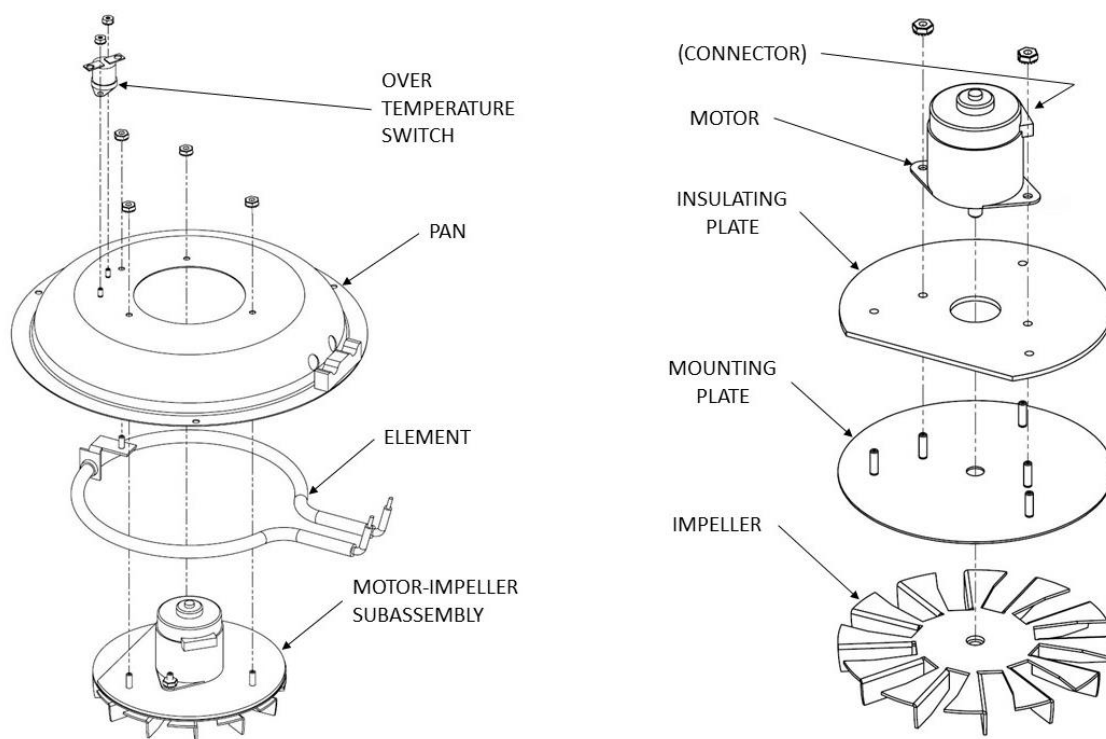


Figure 5.2. Heater Disassembly

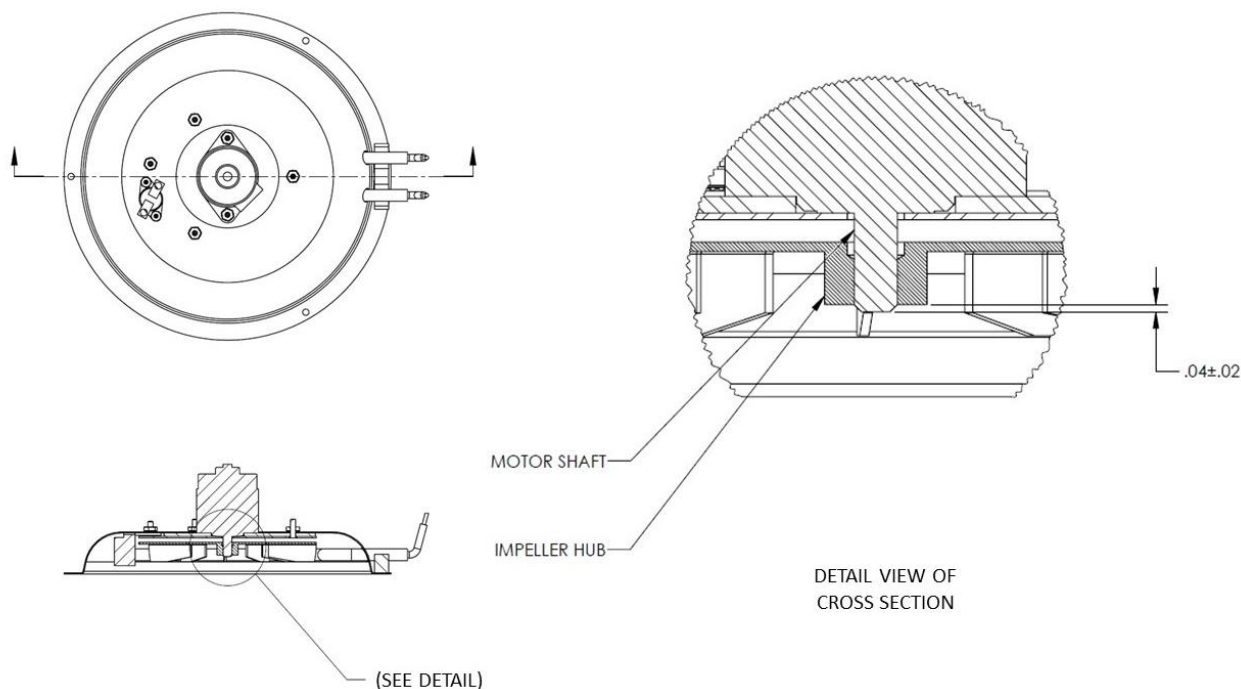


Figure 5.3. Impeller Installation

To replace Heater Impeller Motor with Impeller as Subassembly (Figure 5.2):

- Mount Subassembly in place on Pan. Note orientation: Straight edge on Insulator faces Element mounting bracket.
- Install the (three) nuts holding Motor Mounting Plate to Pan.
- Spin Impeller and check that it spins freely and is not perceptibly bent. A bent Impeller must be replaced.

Heat Element (C1906912)

To remove Element (Figure 5.2):

- Remove the Heater from unit as described above.
- Remove Heater Impeller Motor with Impeller as Subassembly, as described above.
- (Do not disassemble Heater Impeller Motor from Impeller.)
- Remove the nut (6-32 KEPS) holding Element to Pan using size 5/16 (0.313) inch wrench.
- Remove Element from Pan. Handle wires carefully.

To replace Element (Figure 5.2)

- Install Element in Pan, first feeding wire ends through holes in Pan.
- Install the (one) nut holding Element to Pan.
 - There is a little play in the exact location of the mounting bracket on Element.
 - Trial fit the Heater Impeller Motor with Impeller as Subassembly on the Pan. Check that tips of Impeller blades have as much clearance as possible from bracket on Element.
 - Loosen nut, adjust, and tighten nut on the Element bracket if needed.
- Replace Heater Impeller Motor with Impeller as Subassembly, as described above.

RTD Temperature Sensor (902-1167)

The Temperature Sensor is an RTD probe located inside the chamber. (Figure 5.1)

Servicing the Sensor requires access to the Heater but does not require removal of Heater.

To remove Sensor:

- Remove Carousel.
- Cut the two nylon wire ties mounting the Sensor. Do not cut the nylon anchors which are fastened to the Chamber ceiling.
- The Sensor will hang freely in the Chamber.
- Access the Heater as described above.
- Pull the Sensor probe through the Grommet by its cable.
- Disconnect the Sensor wiring at the Main Board. (Figure 3.4)
- Remove Sensor.

To replace Sensor:

- Reverse removal process.
- Re-cover the Heater as described above.

Over-Temperature Switch (926-0021)

To remove Over-Temperature Switch:

- Access the Heater, as described above. It is not necessary to remove the Heater.
- Locate the Switch. (Figure 5.2)
- Disconnect both wires at push-on terminals. (Do not pull directly on wires.)
- Remove nuts (4-40 KEPS, quantity 2) using size 1/4 (0.250) inch open end wrench.
- Remove Switch.

To replace Switch:

- Reverse removal process.
- There is no required polarity for connection of wires, i.e., either of two wires can connect to either of two terminals.
- Re-cover the Heater as described above.

Section 6. Door Seal Service Procedures

➤ Door Seal Replacement

Door Seal (D1906945)

- Lift the door off the hinges. Lay door on its exterior face on a well-padded surface. The door must be primarily supported on its front metal face, not on the glass.
- Loosen and remove screws (6-32 x 1/4" Pan Head SEMS Phillips, quantity 10) from the inner door panel. Remove the inner door panel (with perimeter seal) from the door assembly. Do not disturb the glass window or surrounding insulation.
- When handling the inner door panel, beware of sharp sheet metal edges and corners. Protective gloves are recommended.
- The seal is retained by a perimeter skirt which encompasses the four outer edges of the panel. Remove the seal by stretching it at one corner, then working around the perimeter to pull it off the panel edge.
- To replace the seal, first check orientation as shown in Figure 6.1. The expanding bellows portion of the seal must face up, farthest from the window. Insert one corner of the door panel into the opening formed by the perimeter skirt. Stretch the seal, working around the perimeter to fit it back on to the panel edge.
- Ensure the seal is completely and evenly seated around the panel perimeter.
- Replace the panel in the door assembly, checking that the seal fits inside the outer perimeter of the entire door frame, and that insulation is not pinched between the panel and glass window.
- Replace all (10) screws loosely, then gradually tighten them in an alternating pattern to achieve even loading on the glass window.
- Reinstall the door on hinges.

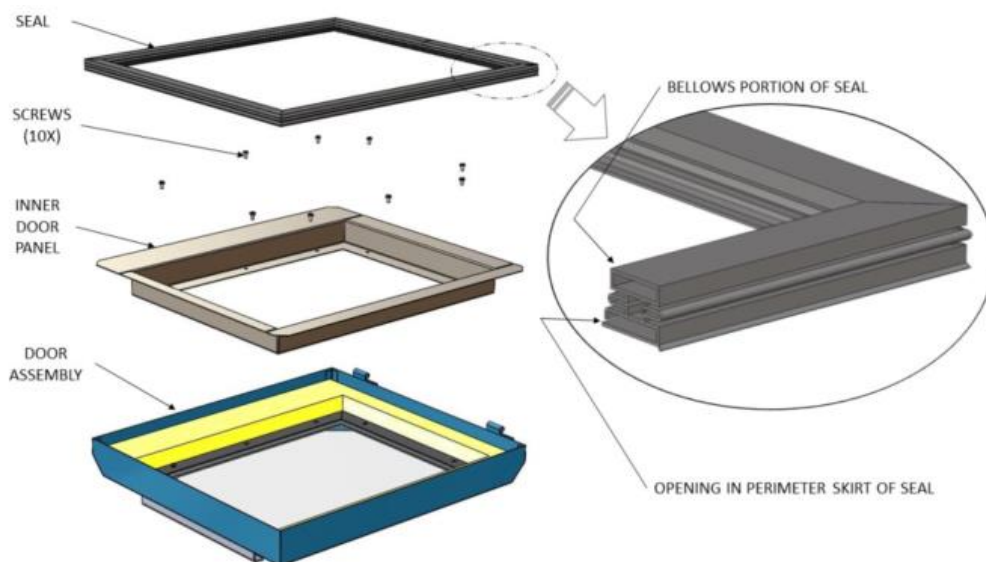


Figure 6.1. Door Seal

Section 7. Engineering Menu

The Engineer Settings Menu is accessible within in the Procleix RES User Interface via the Admin Settings.

- Accessing the Menu
- Upload Firmware
- Development Programs
- Calibrate Temperature
- Reset Engineer Password
- TCIP/IP Setup
- Select Language
- Assay List
- Factory Menu

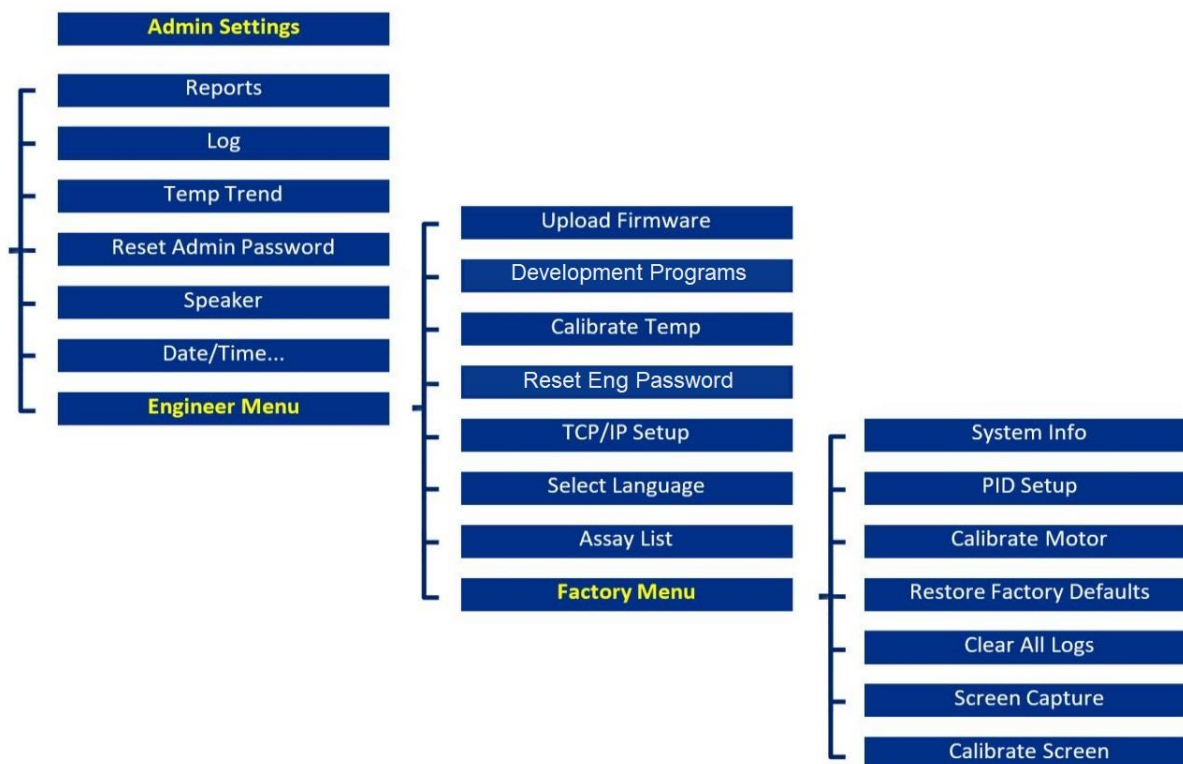


Figure 7.1. RES Management Screen Tree (Summary)

Accessing the Engineering Menu

From the Home Screen (Figure 7.2) press the gears icon on the lower right corner to launch the Admin Settings. (Figure 7.3)

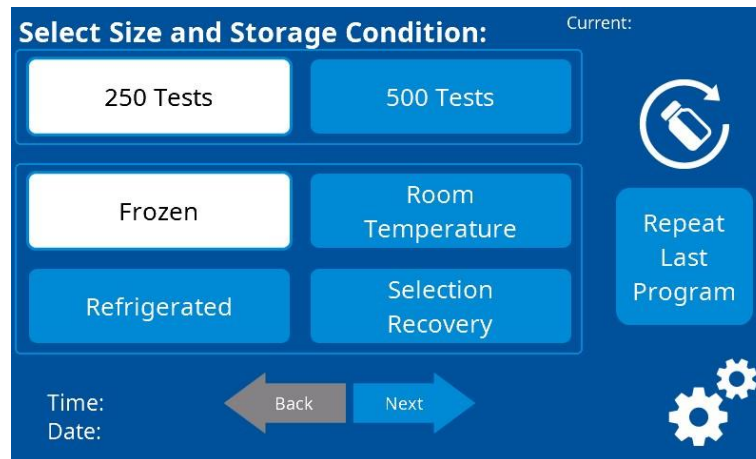


Figure 7.2. Main Home Screen

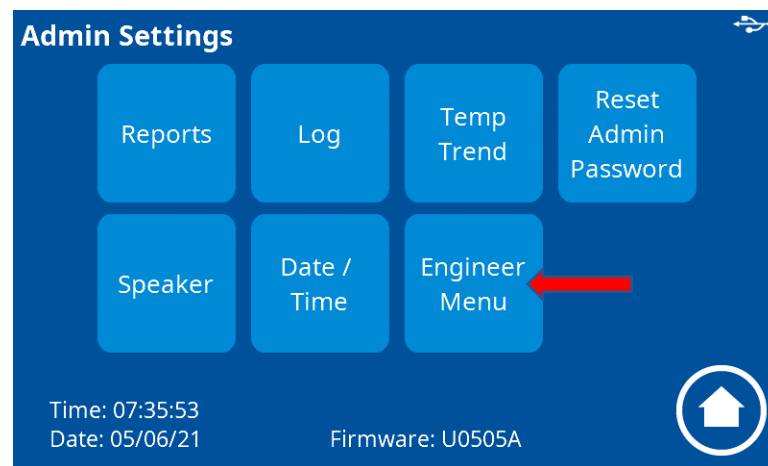


Figure 7.3. Admin Settings Menu

Note: For the Admin Settings Menu functions please refer to *Procleix Reagent Equilibration System Operator's Manual*.

Press the Engineer Menu button to launch the Engineer Settings Menu. (Figure 7.4)

Enter the default password 99999 at the prompt. The Engineering Menu opens.

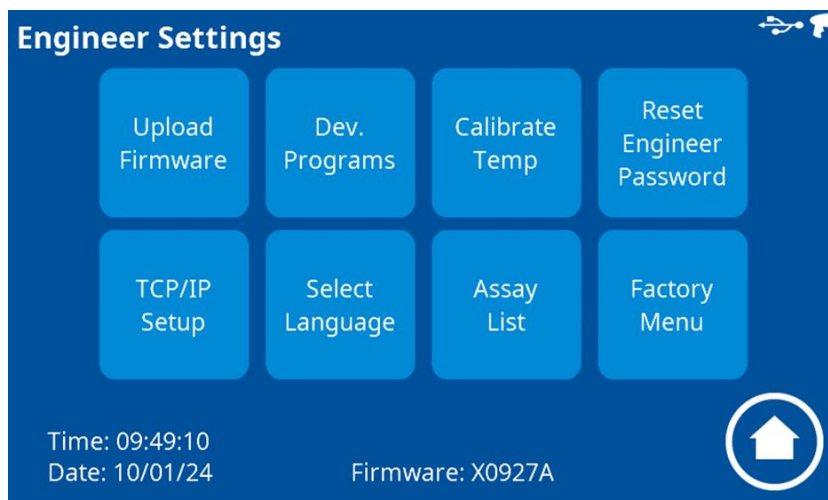


Figure 7.4. Engineer Settings Menu

➤ **Upload Firmware**

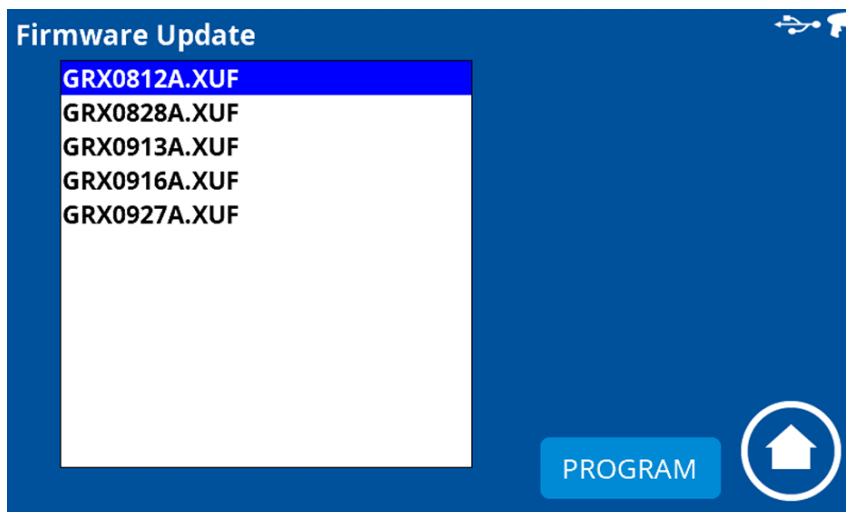


Figure 7.5. Upload Firmware

To download the new firmware file ("xxxxxxx.XUF") provided to a USB:

- Load the USB to the left port on the front of the RES.
- Press "Upload Firmware".
- Screen will display available update file(s).
- Select desired update file, press "PROGRAM".
- Watch progress and follow screen instructions.
- Restart the RES.
- Verify the updated Firmware is listed on the Home Screen.

Note: USB memory drives for use with the RES have the following requirements:

- The USB flash memory drive should be FAT32 formatted with 64GB or less storage capacity.
- The USB flash memory drive must not have pre-loaded software.

- The USB flash memory drive must be dedicated for use with RES. Do not use the designated USB flash memory drive with other file types.

➤ Development Programs

Only create programs that would apply to the preparing the Procleix Reagents, for example a program for the Room Temperature File.

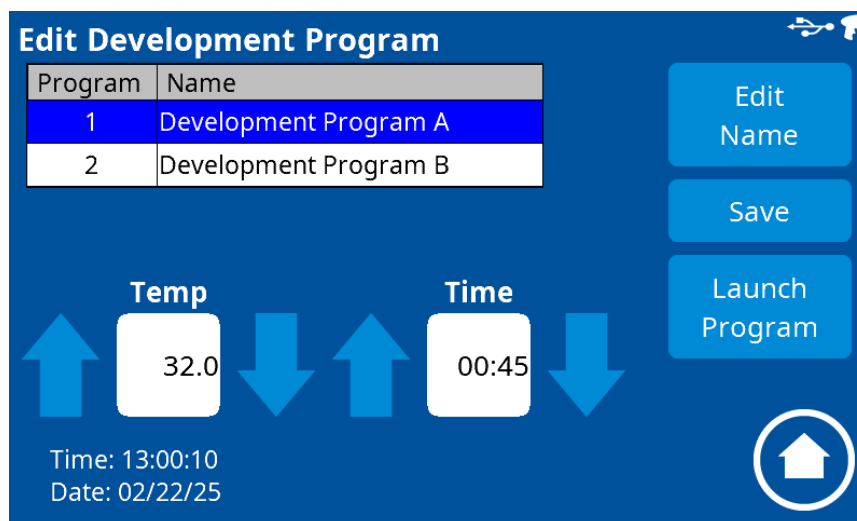


Figure 7.6. Setup Customer Development Programs

- Press “Development Programs”.
- Select one of the Development Programs that is available from the list.
- Edit the name.

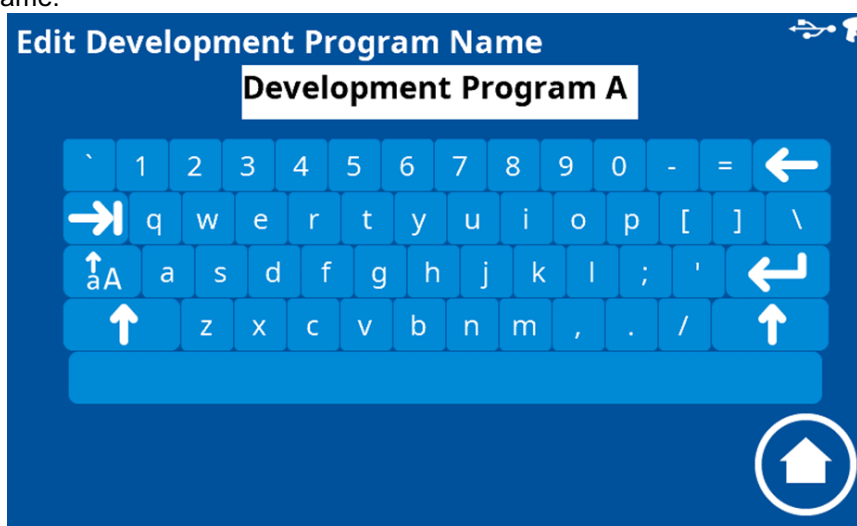


Figure 7.7. Edit Development Program Name

- Select the Temp and the Time.
- Click Save.
- Launch the Program.
- Verify the program is operating per the parameters set.

➤ **Calibrate Temperature**

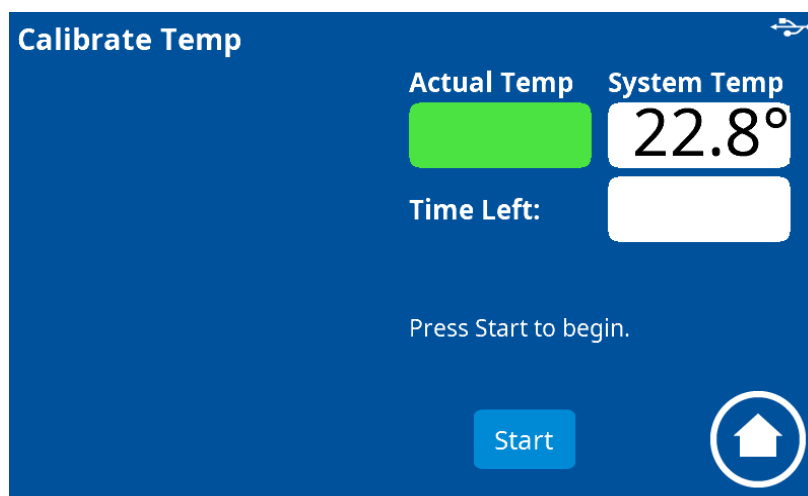


Figure 7.8. Calibrate Temperature

- Temperature calibration for the device may be performed following the onscreen prompts. A calibrated external probe will be required, and it is recommended that the probe be placed near the RES temperature sensor. Note that the carousel will rotate during calibration, so any probe wiring must be temporarily secured out of the way of the carousel rotation. An accuracy of $\pm 0.5^{\circ}\text{C}$ is recommended.

Note: This procedure is performed when a Temperature Verification fails. Refer to Installation and Operational Qualification of the Procleix Reagent Equilibration System (RES) (409773).

- The calibration routine is an automated two-step process. The RES is first set to 32°C for an hour, to ensure that the temperature inside the chamber is stable. The user then enters the temperature from the calibrated reference device and proceeds to the next step. The RES will then achieve 60°C over the next hour and the user will enter the reference temperature after the second hour and save. That will end the calibration routine. The RES will then use these two data points as offsets for the reported temperature going forward, interpolating between them for the range of temperatures.
- Press “Calibrate Temp”.
- Press Start and follow the prompts to begin the calibration.
- The Calibrate Temp at 32°C begins. After the first hour record the Actual Temp of the external probe.
- Upon completion of the calibration at 32°C , the calibration at 60°C for an hour begins.

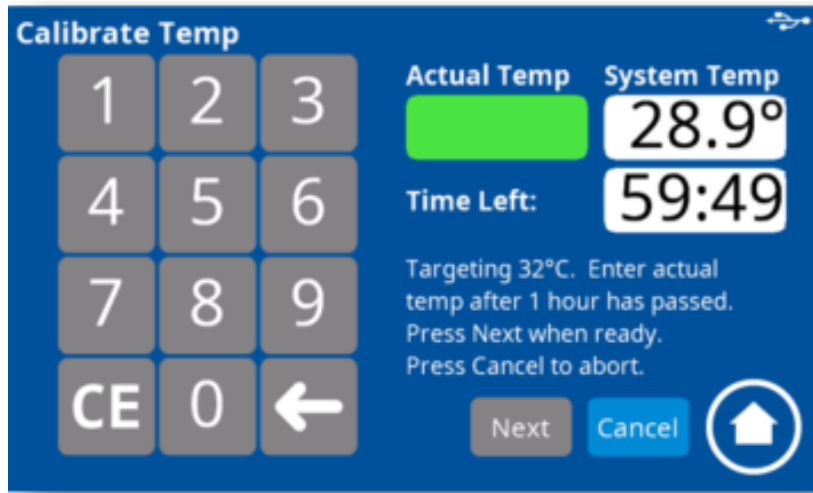


Figure 7.9. Calibration at 32°C

- The Calibrate Temp at 60°C begins. After the first hour, record the Actual Temp of the external probe.

Note: First, it is important that the RES is undisturbed for the process, keeping the door closed and free from drafts.

Second, the hour wait for each setpoint is a minimum. The screen is locked out from input until an hour has passed for each step. It can be longer than an hour if necessary and will not harm the process.

➤ Reset Engineer Password

- The default password is 99999. Reset the password for access to the Engineering Settings screen if needed by pressing “Reset Engineer Password” and follow the prompts.

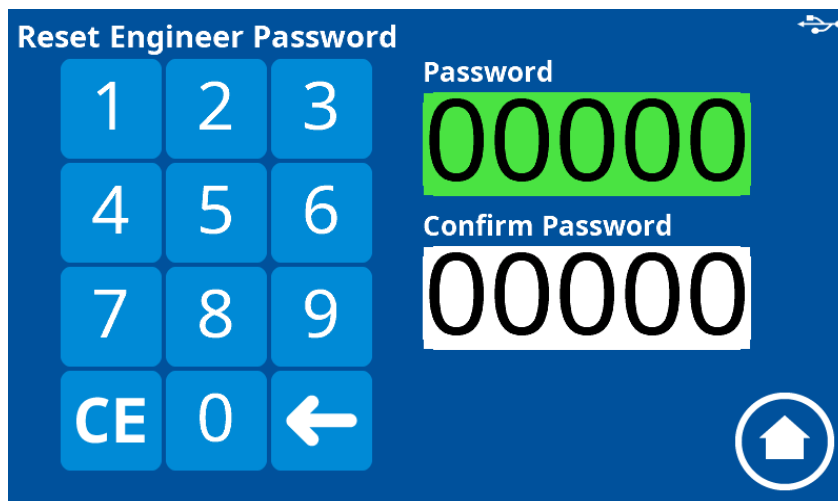


Figure 7.10. Resetting the Engineer Password

Note: There is an active overriding password that will allow Grifols personnel to reset both the Admin or Engineering password if forgotten. The numbers “19053” (Boekel’s zip code, which is on the serial label sticker on the back of the machine) may be input in either the admin or engineering password screens and will allow access beyond, regardless of the default or user-changed passwords for those

menus. Once access is achieved, reset the passwords for the Admin Settings per the customer's requirements and the Engineering Menu password to the default used by all engineers (99999).

➤ **TCP/IP Setup**

TCP/IP Configuration

IP Address: 192 168 000 001

Subnet Mask: 255 255 255 000

Default Gateway: 192 168 000 001

Enable DHCP

Figure 7.11. TCP/IP Setup Menu

- Manage network settings per site.

➤ **Select Language**

- A list of available languages for the GUI are listed in the screen below.

System Language

☒ English ☐ Nihongo

☐ Cestina ☐ Polski

☐ Deutsch ☐ Portugues

☐ Espanol ☐ Slovenscina

☐ Francais ☐ Zhongwen

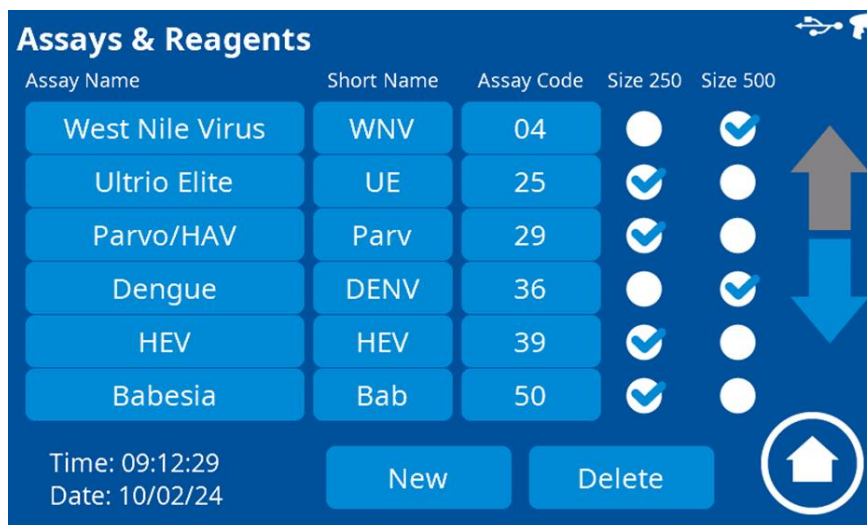
☐ Italiano

Figure 7.12. System Language Choices

- Press "Select Language".
- Select the language appropriate for the end user.
- Press the "Home" button.
- Verify the language in the GUI with the end user.

➤ Assay List

- Press “Assay List”.
- A list of the current Assay Names, Short Name, and Kit size appears in the screen below.



Assay Name	Short Name	Assay Code	Size 250	Size 500
West Nile Virus	WNV	04	☐	☒
Ultrio Elite	UE	25	☒	☐
Parvo/HAV	Parv	29	☒	☐
Dengue	DENV	36	☐	☒
HEV	HEV	39	☒	☐
Babesia	Bab	50	☒	☐

Time: 09:12:29
Date: 10/02/24

New Delete

Figure 7.13. Current Assay & Reagents List

- Scroll down to end of list and press “New” to create a new blank row.
- Highlight the blank space under “Assay Name” and input the new Procleix Assay Name, then push the “Home” button.
- Highlight the blank space for the “Short Name” and input the new Short Name” (no more than 4 characters) then push the “Home” button.
- Highlight the blank space for the “Assay Code” and input the new two-character Procleix Assay Code, then push the “Home” button.

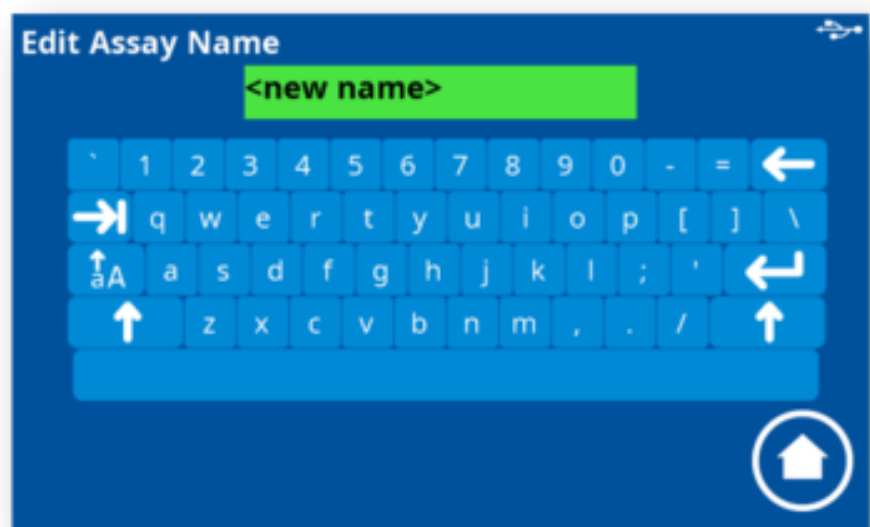


Figure 7.14. Enter New Assay Name

- Select Size 250 then push the “Home” button.

Note: All new Assay reagents will be an Assay Kit Size of 250.

Note: A new Assay should have regulatory approval for location. Assay Name should be name on the Assay Kit form and the “Short Name” and the “Assay Code” provided by Grifols Technical Service to ensure name and code are used universally in all instruments.

- Verify the new Assay is listed in the Assay and Reagents screen during loading reagents.
- As Assays are made obsolete they can be removed from the system via this screen using the Delete function.

➤ **Factory Menu**

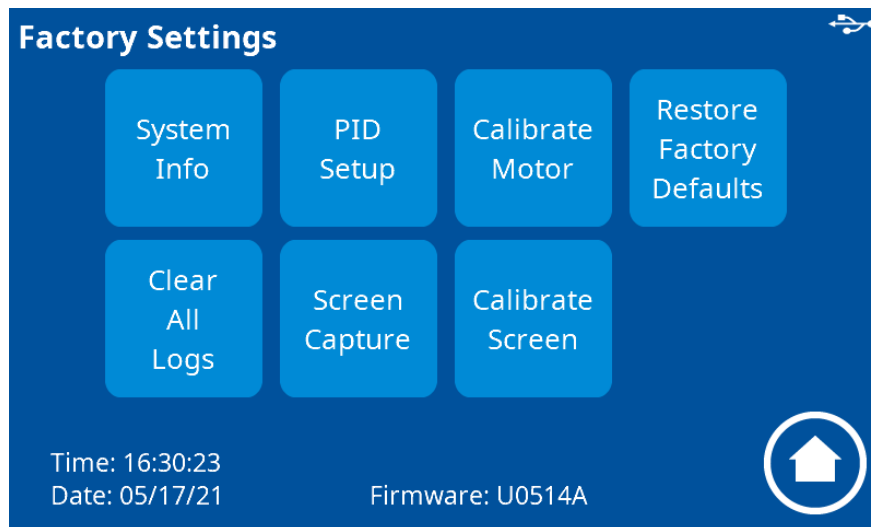


Figure 7.15. Factory Settings Menu

- Press any of the Menu selections to enter and follow the prompts.
- **System Information**

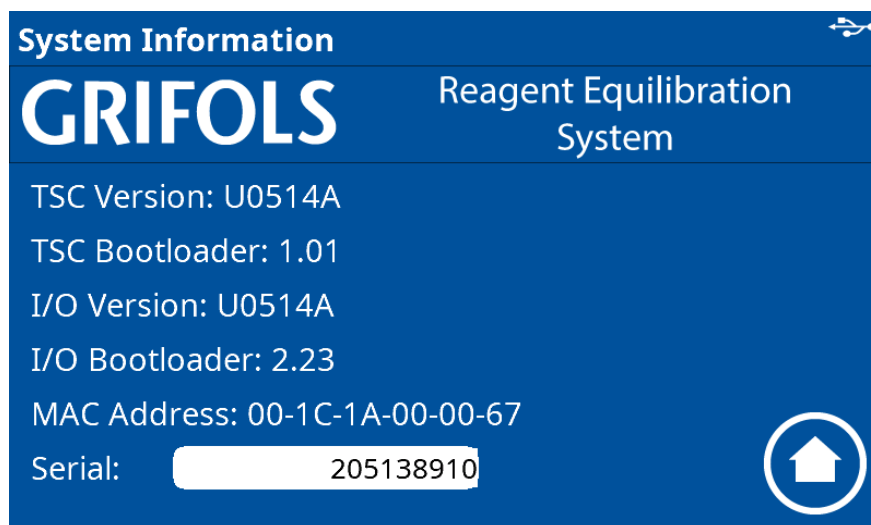


Figure 7.16. RES System Information

- Current system versions are provided.
- Add the RES Serial Number in this screen when installing the instrument if not already entered.

- **PID Setup**

The PID Setup screen features a blue background. On the left is a numeric keypad with buttons for digits 1-9, 0, a 'CE' button, and a back arrow. To the right of the keypad are four input fields: 'Proportional Gain' (highlighted in green with value 40.0000), 'Integral Gain' (value 1.0000), 'Derivative Gain' (value 1500.0000), and 'Time' (value 3.0). A home button icon is in the bottom right corner.

Figure 7.17. PID Setup

Note: These parameters must not be changed in the field. They affect the heating control and have been calculated, thoroughly tested, and verified during product development. Changing any of these may impair or render the heating control completely ineffective until reverted to the shown default parameters.

- **Calibrate Motor**

- Select either the Carousel Motor or the Thermal Fan
- Settings on the “Carousel Motor” and “Thermal Fan” screen must not be changed from the factory defaults. Pressing either of the two buttons brings up the corresponding inputs for that motor, as well as allowing for calibration of the motor RPM using the “Calibrate ...” button on the bottom of the list.

The Carousel Motor Settings screen has a blue background. On the left, under 'Select Motor:', are three buttons: 'Carousel Motor' (white), 'Thermal Fan' (blue), and 'Calibrate Carousel' (blue). To the right are four input fields: 'Gear Ratio' (30), 'Low RPM' (10), 'High RPM' (25), and 'Tolerance' (40). Below these is a radio button labeled 'Reverse Rotation'. At the bottom left, it shows 'Time: 09:13:31' and 'Date: 10/02/24'. A home button icon is in the bottom right corner.

Figure 7.18. Carousel Motor Settings

Select Motor:

Carousel Motor

Thermal Fan

Calibrate Fan

Pulses / Rev 100

Low RPM 500

High RPM 1250

Tolerance 40

☐ Reverse Rotation

Time: 09:13:37
Date: 10/02/24

Figure 7.19. Thermal Fan Settings

- Calibration of either motor is unnecessary unless its output is erratic or unstable. To calibrate, simply enter the menu, press “Start” and allow the motor to warm up for five to ten minutes, noting when the feedback response (RPM and Output) has stabilized. After at least five minutes, the output may be accepted and calibrated.

Motor Calibration

Current RPM 0.0

Current Output 0

Start

Tap Start to continue.

Time: 15:46:10
Date: 07/21/21

Figure 7.20. Motor Calibration

- **Restore Factory Defaults**

- Factory settings may need to be restored.

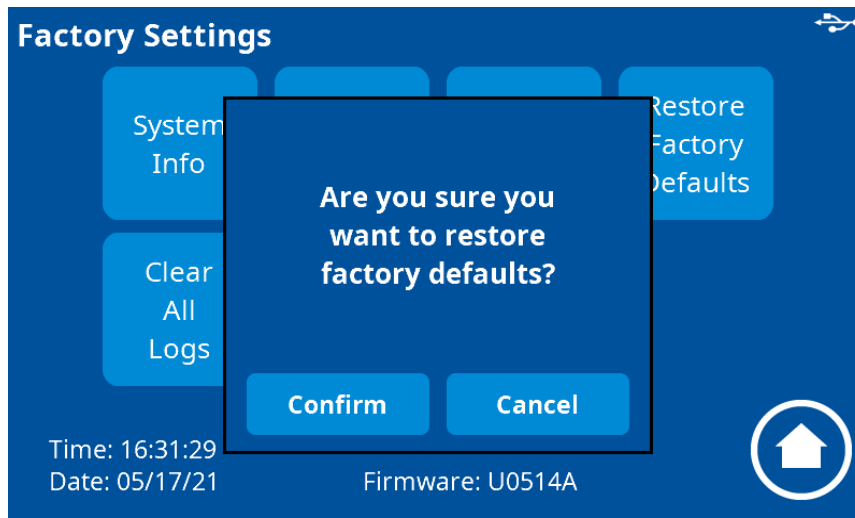


Figure 7.21. Restore Factory Default Settings Confirmation

- To restore all factory default settings select "Confirm".

- **Clear All Logs**

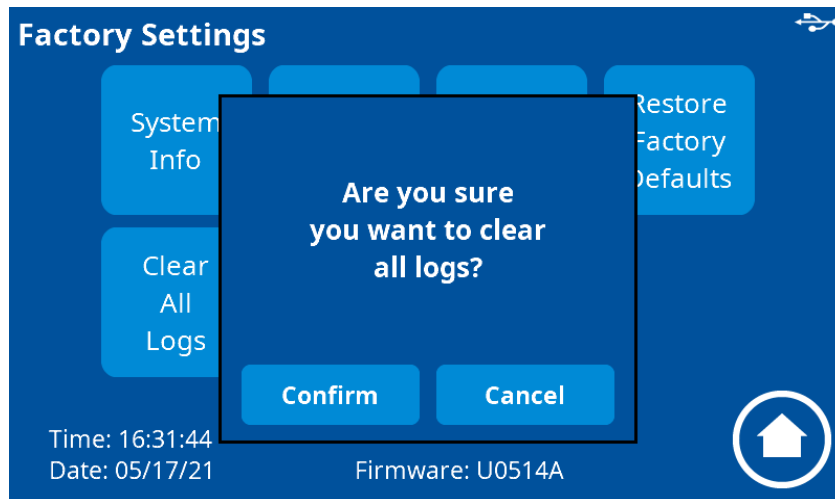


Figure 7.22. Clear All Logs Confirmation

- To clear all stored logs select "Confirm".

Note: Once a 3 month time frame is reached new logs will overwrite oldest existing logs automatically.

- **Screen Capture**

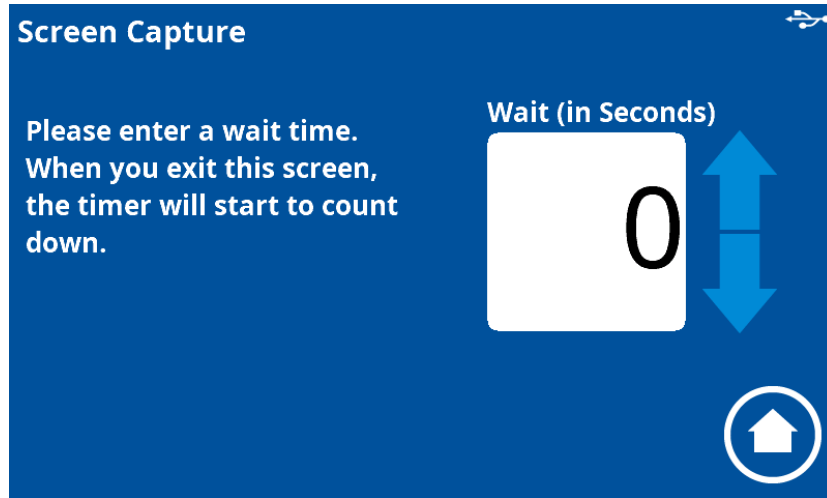
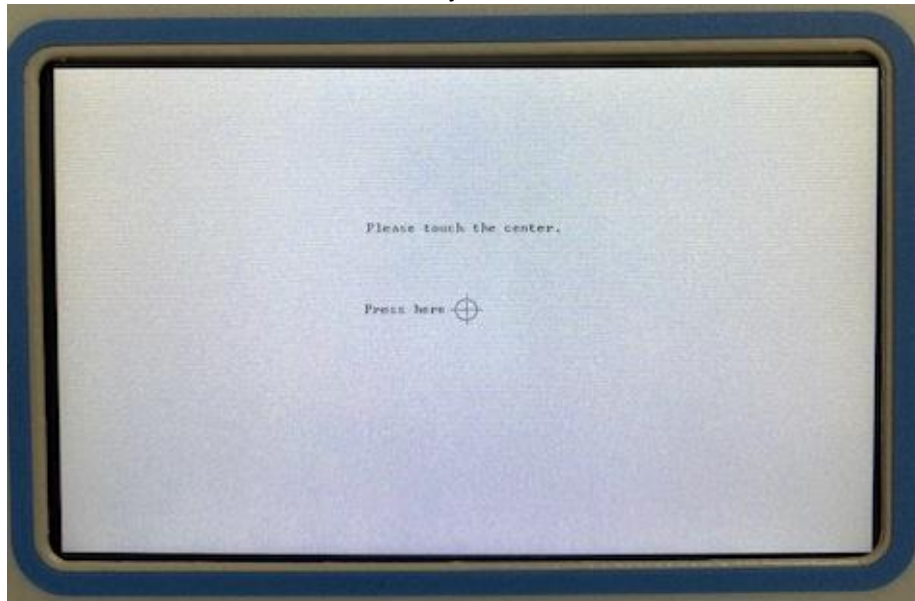


Figure 7.23. Screen Capture View

- Insert USB into the left USB port.
- Set time needed to go back to the desired screen for capture (at least 20 second).
- Click on Home button to navigate to the desired screen for capture.
- Wait for the message for screen capture.
- Remove USB.

- **Calibrate Screen**

- Follow the onscreen prompts by touching the targets as they move around the screen in order to recalibrate the touch location accuracy of the screen.



Section 8. Troubleshooting

Replacing Batteries:

Upon restarting the RES, if the unit will not progress past the screen below and complete powering up, then the internal 4x AA batteries (915-0095) should be removed and reloaded which reboots the main board.



Figure 8.1. Start-up Screen

- Power off the unit and disconnect the AC cord.
- Take off the back panel and locate the 4x AA batteries on the left side.

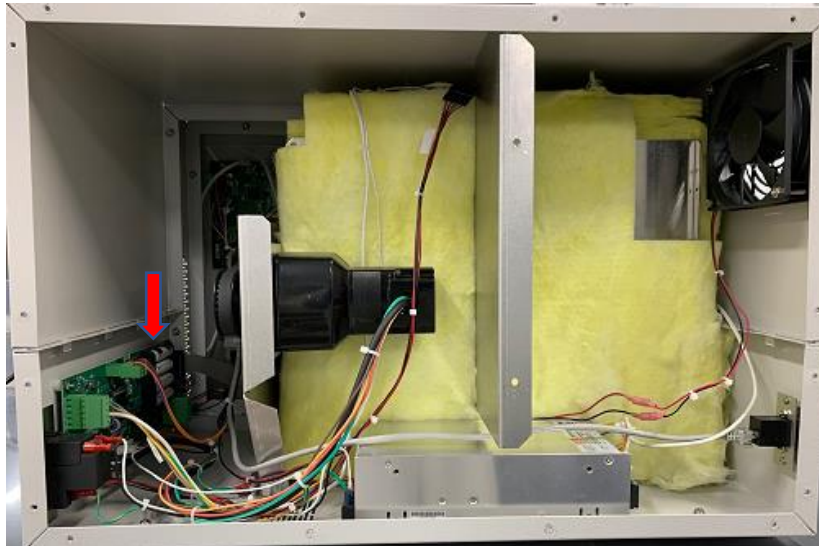


Figure 8.2. Location of Batteries

-
- Remove the batteries for 2 minutes in order to completely turn off the power to the unit. Note the polarity of the batteries as they are removed in order to put them back in the same orientation.



Figure 8.3. Batteries

- Place the batteries back in the unit and power on the RES.
- If the unit still will not power up properly, replace the main board.

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Appendix B. Support Contact Information

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